

Differentiation of Exponential Functions and Applications

1. Find $\frac{dy}{dx}$ for each of the following:

(a) $y = e^x - 3$

(b) $y = 5e^x + 2x$

(c) $y = (e^x + 7)^3$

(d) $y = xe^x$

(e) $y = \frac{e^x}{x}$

(f) $y = \sqrt{2e^x - 5}$

(a) $\frac{dy}{dx} = e^x$ (b) $\frac{dy}{dx} = 5e^x + 2$ (c) $\frac{dy}{dx} = 3(e^x + 7)^2 \cdot e^x$

(d) $\frac{dy}{dx} = x \cdot e^x + e^{x-1} = e^x(x+1)$ (e) $y = \frac{e^x}{x}$
 $\frac{dy}{dx} = \frac{x \cdot e^x - e^x}{x^2} = \frac{e^x(x-1)}{x^2}$ (f) $y = (2e^x - 5)^{\frac{1}{2}}$
 $\frac{dy}{dx} = \frac{1}{2} (2e^x - 5)^{-\frac{1}{2}} \cdot (2e^x)$
 $\frac{dy}{dx} = \frac{e^x}{\sqrt{2e^x - 5}}$

2. If $y = e^{kx}$ is a solution of the equation $\frac{d^2y}{dx^2} + \frac{dy}{dx} - 6y = 0$ find the possible values of the constant k .

$y = e^{kx}$
 $\frac{dy}{dx} = k \cdot e^{kx}$
 $\frac{d^2y}{dx^2} = k^2 \cdot e^{kx}$

$k^2 \cdot e^{kx} + k \cdot e^{kx} - 6e^{kx} = 0$
 $e^{kx} (k^2 + k - 6) = 0$
 $\neq 0 \quad (k+3)(k-2) = 0$
 $k = -3 \text{ or } 2$

Differentiation of Exponential Functions and Applications continued

3. Find the equation of the tangent and the normal to the curve $y = 2e^{3x} - 1$ at the point where $x = 0$.

$$x=0, y = 2e^{3(0)} - 1 = 1 \rightarrow (0, 1)$$

$$y = 2e^{3x} - 1$$

$$y' = 2e^{3x} \cdot 3 = 6e^{3x}$$

$$y' = 6e^{3x}$$

$$x=0, m_T = 6e^{3(0)} = 6$$

$$m_N = -\frac{1}{6}$$

Tangent: $y - 1 = 6(x - 0)$
 $y = 6x + 1$ ✓

Normal: $y - 1 = -\frac{1}{6}(x - 0)$
 $y = -\frac{x}{6} + 1$ ✓

4. The price \$p\$ of a lipstick is given by $p = 10e^{-x}$ for $0 < x < 2$ where x is the number of lipsticks (in thousands) that can be sold in a week.

- (a) Revenue is defined as the total dollar amount received for the sale of x lipsticks. Find the price at which the weekly revenue will be a maximum.
(b) Find the maximum weekly revenue. x is measured in thousands

(a) Revenue: $R(x) = (1000x)(10e^{-x}) = 10000xe^{-x}, 0 < x < 2$

$$R'(x) = 10000x \cdot e^{-x}(-1) + 10,000e^{-x}$$

$$R'(x) = 10,000e^{-x}(1-x)$$

$$R'(x) = 0 \Rightarrow 10,000e^{-x}(1-x) = 0$$

$$1-x = 0$$

$$x = 1$$

Nature:

$$R'(x) = 10,000e^{-x}(1-x)$$

x	$\frac{1}{2}$	1	1.5
$R'(x)$	+	0	-

Maximum Revenue at $x = 1$

$$\text{Price, } P = 10e^{-x} = 10e^{-1} = \frac{10}{e} \approx \$3.68$$

(b) Maximum Revenue, $R(x) = 10000xe^{-x}$
 $x=1, R(1) = 10000 \cdot 1 \cdot e^{-1} = \frac{10000}{e} \approx \$3678.79/\text{week}$

Differentiation of Exponential Functions and Applications continued

5. Consider the function $y = xe^{-x}$.

(a) Find the coordinates of the stationary point and determine its nature.

(b) Find the value of x for which:

(i) $y > 0$ (ii) $\frac{dy}{dx} > 0$ (iii) $\frac{d^2y}{dx^2} > 0$

(c) Given that $\lim_{x \rightarrow \infty} \frac{x}{e^x} = 0$ write down the equation of the horizontal asymptote of the function $y = xe^{-x}$.

(d) Sketch the graph of the function $y = xe^{-x}$.

(c) $\lim_{x \rightarrow \infty} \frac{x}{e^x} = 0$
 $\therefore y = 0$ (x -axis) is the H.A.

(a) $y = xe^{-x}$

$y' = x \cdot e^{-x}(-1) + e^{-x} \cdot 1$

$y' = e^{-x}(1-x)$

S.P.: $y' = 0$

$e^{-x}(1-x) = 0$

$\neq 0 \quad x = 1, y = 1 \cdot e^{-1} = \frac{1}{e}$

$(1, \frac{1}{e})$ is the stationary point

relative: $y' = e^{-x}(1-x)$

$y'' = e^{-x}(-1) + (1-x)e^{-x} \cdot (-1)$

$= e^{-x}(-1-1+x)$

$y'' = e^{-x}(x-2)$

at $x=1, y'' = e^{-1}(1-2) = -\frac{1}{e} < 0$

Maximum T.P. at $(1, \frac{1}{e})$

(b) $y = xe^{-x}$

(i) $y > 0$

$\frac{x}{e^x} > 0$

$x > 0$

(ii) $\frac{dy}{dx} > 0$

$e^{-x}(1-x) > 0$

$1-x > 0$

$-x > -1$

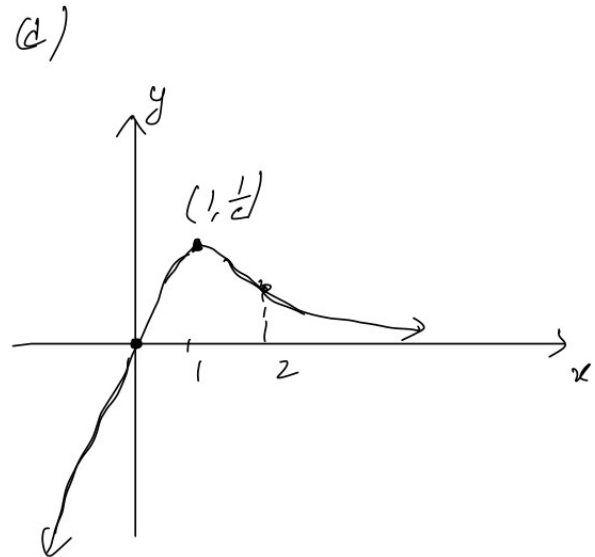
$x < 1$

(iii) $\frac{d^2y}{dx^2} > 0$

$e^{-x}(x-2) > 0$

$x-2 > 0$

$x > 2$



Differentiation of Logarithmic Functions

6. Find $\frac{dy}{dx}$:

(a) $y = \ln 3x$

(c) $y = \ln(x^2 - 3x + 2)$

(e) $y = x \ln x$

(b) $y = \ln(x^2 - 5)$

(d) $y = \ln(1 - x^3)$

(f) $y = e^x \ln x$

(a) $y = \ln 3x$
 $y = \ln 3 + \ln x$
 $y' = 0 + \frac{1}{x} = \frac{1}{x}$

(d) $y = \ln(1 - x^3)$
 $y' = \frac{-3x^2}{1 - x^3}$

(b) $y = \ln(x^2 - 5)$
 $y' = \frac{2x}{x^2 - 5}$

(c) $y = x \cdot \ln x$
 $y' = x \cdot \frac{1}{x} + \ln x \cdot 1 = 1 + \ln x$

(c) $y = \ln(x^2 - 3x + 2)$
 $y' = \frac{2x - 3}{x^2 - 3x + 2}$

(f) $y = e^x \cdot \ln x$
 $y' = e^x \cdot \frac{1}{x} + \ln x \cdot e^x$
 $y' = e^x \left(\frac{1}{x} + \ln x \right)$

7. Differentiate with respect to x :

(a) $\ln x^9$

(c) $\ln\left(\frac{x}{2}\right)$

(e) $\ln\sqrt{x+5}$

(b) $\ln(5x-7)^8$

(d) $\ln\left(\frac{7x-6}{x+8}\right)$

(f) $\ln\sqrt{x^2-1}$

(a) $y = \ln x^9$
 $y = 9 \ln x$
 $y' = \frac{9}{x}$

(d) $y = \ln\left(\frac{7x-6}{x+8}\right)$
 $y = \ln(7x-6) - \ln(x+8)$
 $y' = \frac{7}{7x-6} - \frac{1}{x+8}$

(b) $y = \ln(5x-7)^8$
 $y = 8 \ln(5x-7)$
 $y' = 8 \cdot \frac{5}{5x-7} = \frac{40}{5x-7}$

(e) $y = \ln\sqrt{x+5} = \ln(x+5)^{\frac{1}{2}} = \frac{1}{2} \ln(x+5)$
 $y' = \frac{1}{2} \cdot \frac{1}{x+5} = \frac{1}{2(x+5)}$

(c) $y = \ln\left(\frac{x}{2}\right)$
 $y = \ln x - \ln 2$
 $y' = \frac{1}{x} - 0$
 $y' = \frac{1}{x}$

(f) $y = \ln\sqrt{x^2-1}$
 $= \ln(x^2-1)^{\frac{1}{2}}$
 $y = \frac{1}{2} \ln(x^2-1)$
 $y' = \frac{1}{2} \cdot \frac{2x}{x^2-1} = \frac{x}{x^2-1}$

Differentiation of Logarithmic Functions continued

8. If $f(x) = (x+1)\ln x$ find $f'(1)$ and $f''(1)$.

$$f'(x) = (x+1) \cdot \frac{1}{x} + \ln x \cdot 1 = 1 + \frac{1}{x} + \ln x$$

$$f'(x) = 1 + \frac{1}{x} + \ln x, \quad x=1, \quad f'(1) = 1 + \frac{1}{1} + \ln(1) = 2$$

$$\therefore \underline{\underline{f'(1) = 2}}$$

$$f''(x) = 0 + \left(-\frac{1}{x^2}\right) + \frac{1}{x}$$

$$f''(x) = -\frac{1}{x^2} + \frac{1}{x}$$

$$x=1, \quad f''(1) = -\frac{1}{1^2} + \frac{1}{1} = -1 + 1 = 0, \quad \therefore \underline{\underline{f''(1) = 0}}$$

9. Use the Change of Base result to differentiate:

(a) $\log_7 x$

$$y = \log_7 x = \frac{\ln x}{\ln 7}$$

$$y' = \frac{1}{\ln 7} \cdot \frac{1}{x}$$

$$y' = \frac{1}{x \ln 7}$$

(b) $\log_{10}(3x-5)$

$$y = \log_{10}(3x-5)$$

$$y = \frac{\ln(3x-5)}{\ln 10}$$

$$y' = \frac{1}{\ln 10} \cdot \frac{3}{3x-5}$$

$$y' = \frac{3}{(3x-5) \ln 10}$$

(c) $\log_5(x^2+1)$

$$y = \log_5(x^2+1) = \frac{\ln(x^2+1)}{\ln 5}$$

$$y' = \frac{1}{\ln 5} \cdot \frac{2x}{x^2+1}$$

$$y' = \frac{2x}{(x^2+1) \ln 5}$$

10. (a) Find the derivative of a^x using the identity $a^x = e^{\ln a^x} = e^{x \ln a}$.

(b) Hence find the derivative of:

(i) 7^x

$$(i) \quad \underline{\underline{7^x \ln 7}}$$

(ii) 9^{2x-5}

$$= 9^{2x-5} \cdot \ln 9 \cdot 2$$

$$= \underline{\underline{2 \ln 9 \cdot 9^{2x-5}}}$$

Integration Involving Exponential Functions and Applications

11. Find the following indefinite integrals:

~~(d)~~ $\int 6xe^{x^2} dx$

~~(e)~~ $\int xe^{x^2} dx$

~~(f)~~ $\int 6x^2e^{x^3} dx$

~~(g)~~ $\int (x+1)e^{x^2+2x} dx$

~~(h)~~ $\int \frac{2}{\sqrt{x}e^{\sqrt{x}}} dx$

~~(i)~~ $\int \sqrt{e^x} dx$

(d) $\int 6x \cdot e^{x^2} dx = 3 \int 2x \cdot e^{x^2} dx = 3 \cdot e^{x^2} + C$

(e) $\frac{1}{2} \int 2x \cdot e^{x^2} dx = \frac{1}{2} e^{x^2} + C$

(f) $\int 6x^2 \cdot e^{x^3} dx = 2 \int 3x^2 \cdot e^{x^3} dx = 2 \cdot e^{x^3} + C$

(g) $\frac{1}{2} \int 2(x+1) e^{x^2+2x} dx = \frac{1}{2} e^{x^2+2x} + C$

(h) $\int \frac{2}{\sqrt{x} e^{\sqrt{x}}} dx = -2 \int \frac{1}{2\sqrt{x}} \cdot e^{-\sqrt{x}} dx = -4 \cdot e^{-\sqrt{x}} + C = \frac{-4}{e^{\sqrt{x}}} + C$

(i) $\int \sqrt{e^x} dx = \int (e^x)^{\frac{1}{2}} dx = \int e^{\frac{1}{2}x} dx = 2 \cdot e^{\frac{1}{2}x} + C$

Integration Involving Exponential Functions and Applications continued

12. Evaluate each of the following definite integrals:

$$(a) \int_1^2 3e^{4x} dx$$

$$(b) \int_0^1 4x^3 e^{2x^4} dx$$

$$(c) \int_1^4 \frac{e^{\sqrt{x}}}{\sqrt{x}} dx$$

$$(d) \int_1^3 3e^{2x} (e^{-2x} - 1) dx$$

$$(e) \int_{-1}^1 \frac{dx}{e^{2-3x}}$$

$$(a) \int_1^2 3e^{4x} dx = 3 \left[\frac{e^{4x}}{4} \right]_1^2 = \frac{3}{4} [e^8 - e^4]$$

$$(b) \frac{1}{2} \int_0^1 2 \cdot 4x^3 \cdot e^{2x^4} dx = \frac{1}{2} [e^{2x^4}]_0^1 = \frac{1}{2} [e^2 - 1]$$

$$(c) 2 \int_1^4 \frac{1}{2\sqrt{x}} \cdot e^{\sqrt{x}} dx = 2 [e^{\sqrt{x}}]_1^4 = 2[e^2 - e]$$

$$(d) \int_1^3 3e^{2x} (e^{-2x} - 1) dx = \int_1^3 (3 - 3e^{2x}) dx = \left[3x - \frac{3e^{2x}}{2} \right]_1^3 = \left(9 - \frac{3e^6}{2} \right) - \left(3 - \frac{3e^2}{2} \right) \\ = 6 - \frac{3}{2}e^6 + \frac{3}{2}e^2 \\ = 6 - \frac{3}{2}(e^6 - e^2)$$

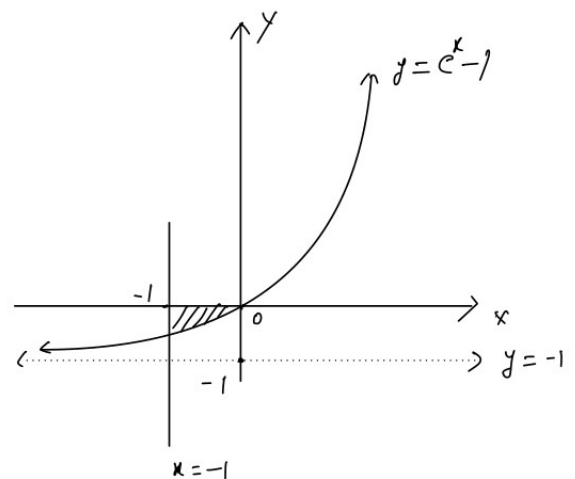
$$(e) \int_{-1}^1 \frac{dx}{e^{2-3x}} = \int_{-1}^1 (e^{2-3x})^{-1} dx \\ = \int_{-1}^1 e^{3x-2} dx = \left[\frac{e^{3x-2}}{3} \right]_{-1}^1 \\ = \frac{1}{3} [e - e^{-5}]$$

13. Sketch the graph of $y = e^x - 1$ and hence find the area bounded by the curve, the x -axis and the line $x = -1$.

$$\text{Area, } A = \left| \int_{-1}^0 (e^x - 1) dx \right| \\ = \left| [e^x - x]_{-1}^0 \right| \\ = \left| (1 - 0) - (e^{-1} + 1) \right| \\ = \left| 1 - \frac{1}{e} - 1 \right|$$

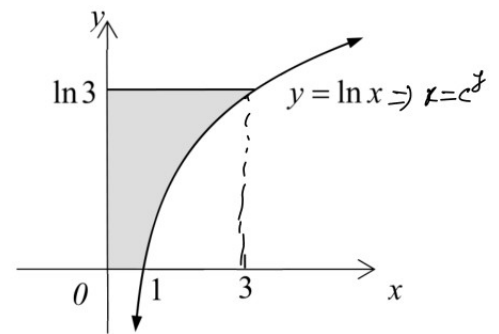
$$A = \left| -\frac{1}{e} \right|$$

$$A = \frac{1}{e} \text{ units}^2$$



Integration Involving Exponential Functions and Applications continued

14. The diagram on the right shows the region bounded by the curve $y = \ln x$, the coordinate axes and the line $y = \ln 3$.



- (a) Find the area of the shaded region.
- (b) Hence find the exact value of $\int_1^3 \ln x \, dx$.
- (c) Find the exact volume of the solid formed by rotating the shaded region about the y-axis.

(a) Area, $A = \int_0^{\ln 3} e^y \, dy$

$$= [e^y]_0^{\ln 3}$$

$$= e^{\ln 3} - e^0$$

$$= 3 - 1$$

$$A = 2 \text{ units}^2$$

(b) $\int_1^3 \ln x \, dx = 3 \cdot \ln 3 - 2$

(c) Volume, $V_y = \pi \int_0^{\ln 3} (e^y)^2 \, dy$

$$= \pi \int_0^{\ln 3} e^{2y} \, dy$$

$$= \pi \left[\frac{e^{2y}}{2} \right]_0^{\ln 3}$$

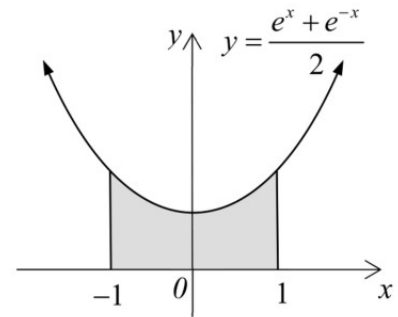
$$= \frac{\pi}{2} [e^{2 \ln 3} - e^{2(0)}]$$

$$= \frac{\pi}{2} [9 - 1]$$

$$V = 4\pi \text{ units}^3$$

Integration Involving Exponential Functions and Applications continued

15. (a) Find the area of the region bounded by the curve $y = \frac{e^x + e^{-x}}{2}$, the lines $x = -1$ and $x = 1$ and the x -axis as shown in the diagram on the right.



- (b) Find the volume of the solid of revolution obtained when this region is rotated about the x -axis.

$$\textcircled{a} \quad \text{Area, } A = \int_{-1}^1 \left(\frac{e^x + e^{-x}}{2} \right) dx = \overset{\text{symmetry}}{\cancel{2}} \cdot \int_0^1 \left(\frac{e^x + e^{-x}}{2} \right) dx$$

$$= \left[e^x + \frac{e^{-x}}{-1} \right]_0^1 = \left[e^x - e^{-x} \right]_0^1 = (e - e^{-1}) - (1 - 1)$$

$$A = \left(e - \frac{1}{e} \right) \text{ units}^2$$

$$\textcircled{b} \quad \text{Volume, } V_x = \pi \int_{-1}^1 \left(\frac{e^x + e^{-x}}{2} \right)^2 dx = \overset{\text{symmetry}}{\cancel{2}} \pi \int_0^1 \left(\frac{e^{2x} + 2 + e^{-2x}}{4} \right) dx$$

$$V_x = \frac{\pi}{2} \left[\frac{e^{2x}}{2} + 2x + \frac{e^{-2x}}{-2} \right]_0^1$$

$$= \frac{\pi}{4} \left[e^{2x} + 4x - e^{-2x} \right]_0^1$$

$$= \frac{\pi}{4} \left[(e^2 + 4 - e^{-2}) - (1 + 0 - 1) \right]$$

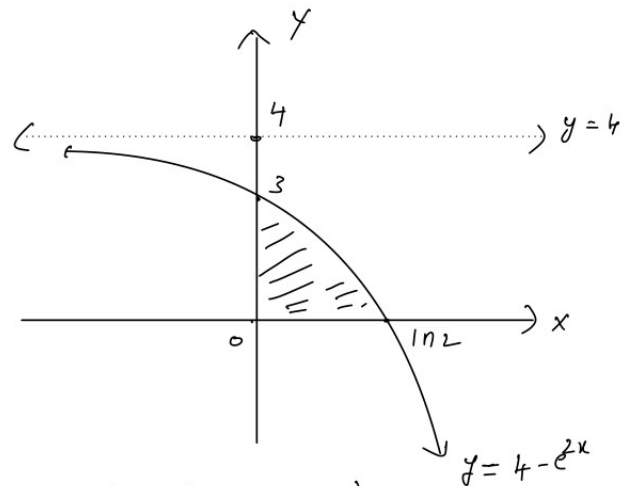
$$V_x = \frac{\pi}{4} (e^2 + 4 - e^{-2}) \text{ units}^3$$

Integration Involving Exponential Functions and Applications continued

16. (a) Find the x -intercept of the curve $y = 4 - e^{2x}$.
- (b) Hence find the area of the region bounded by the curve $y = 4 - e^{2x}$ and the coordinate axes.
- (c) Find the volume of the solid of revolution obtained when the region in part (b) is rotated about the x -axis.



(a) $x \rightarrow \text{int: } y = 0$
 $4 - e^{2x} = 0$
 $4 = e^{2x}$
 $\Rightarrow 2x = \ln 4$
 $x = \frac{\ln 4}{2}$
 $x = \ln 2$



(b) Area, $A = \int_0^{\ln 2} (4 - e^{2x}) dx$
 $= \left[4x - \frac{e^{2x}}{2} \right]_0^{\ln 2} = \left(4 \ln 2 - \frac{1}{2} e^{2 \ln 2} \right) - \left(4(0) - \frac{1}{2} e^{2(0)} \right)$
 $= 4 \ln 2 - \frac{1}{2} (4^2) + \frac{1}{2}$
 $A = \left(4 \ln 2 - \frac{3}{2} \right) \text{ units}^2$

(c) Volume, $V_x = \pi \int_0^{\ln 2} (4 - e^{2x})^2 dx = \pi \int_0^{\ln 2} (16 - 8e^{2x} + e^{4x}) dx$
 $V_x = \pi \left[16x - 8 \cdot \frac{e^{2x}}{2} + \frac{e^{4x}}{4} \right]_0^{\ln 2}$
 $= \pi \left[\left(16 \ln 2 - 4 e^{2 \ln 2} + \frac{1}{4} e^{4 \ln 2} \right) - \left(16 \cdot 0 - 4 e^{2 \cdot 0} + \frac{1}{4} e^{4 \cdot 0} \right) \right]$
 $= \pi \left[\left(16 \ln 2 - 16 + 4 \right) - \left(-4 + \frac{1}{4} \right) \right]$
 $= \pi \left[16 \ln 2 - 12 + \frac{15}{4} \right] = \pi \left[16 \ln 2 - 8 \frac{1}{4} \right] \text{ units}^3$

Indefinite Integrals Involving Logarithmic Functions

17. Find indefinite integrals for each of the following:

(g) $\frac{2x-3}{2x^2-6x+1}$

(h) $\frac{ax+b}{ax^2+2bx+c}$

(i) $\frac{4-2x}{4x-x^2}$

(s) $\frac{e^x - e^{-x}}{e^x + e^{-x}}$

(t) $\frac{1}{e^{-2x} + 1}$

(u) $\frac{3 \sin x}{6 + 5 \cos x}$

(v) $\frac{\sec^2 2x}{1 - 3 \tan 2x}$

(g) $\frac{1}{2} \int \frac{2(2x-3)}{2x^2-6x+1} dx = \frac{1}{2} \ln |2x^2-6x+1| + C$

(h) $\frac{1}{2} \int \frac{2(ax+b)}{ax^2+2bx+c} dx = \frac{1}{2} \ln |ax^2+2bx+c| + C$

(i) $\int \frac{4-2x}{4x-x^2} dx = \ln |4x-x^2| + C$

(s) $\int \frac{e^x - e^{-x}}{e^x + e^{-x}} dx = \ln (e^x + e^{-x}) + C$

(t) $\int \frac{1}{e^{-2x} + 1} dx = \int \frac{1}{\frac{1}{e^{2x}} + 1} dx = \frac{1}{2} \int \frac{2e^{2x}}{1+e^{2x}} dx$
 $= \frac{1}{2} \ln (1+e^{2x}) + C$

(u) $\int \frac{3 \sin x}{6+5 \cos x} dx$

$= \frac{3}{-5} \int \frac{-5 \sin x}{6+5 \cos x} dx$

$= -\frac{3}{5} \ln |6+5 \cos x| + C$

(v) $\frac{1}{-6} \int \frac{-6 \sec^2 2x}{1-3 \tan 2x} dx$

$= -\frac{1}{6} \ln |1-3 \tan 2x| + C$

18. Show that $\frac{1}{(x+1)(x+2)} = \frac{1}{x+1} - \frac{1}{x+2}$. Hence find $\int \frac{dx}{(x+1)(x+2)}$.

$$\begin{aligned} \int \frac{dx}{(x+1)(x+2)} &= \int \left(\frac{1}{x+1} - \frac{1}{x+2} \right) dx \\ &= \ln |x+1| - \ln |x+2| + C \\ &= \ln \left| \frac{x+1}{x+2} \right| + C \end{aligned}$$

Indefinite Integrals Involving Logarithmic Functions continued

19. (a) Show that $\frac{x^2+1}{x+1} = x-1 + \frac{2}{x+1}$. $\underline{RHS} = x-1 + \frac{2}{x+1} = \frac{x^2-1+2}{x+1} = \frac{x^2+1}{x+1} = \underline{LHS}$

(b) Find the equation of the curve for which $\frac{dy}{dx} = \frac{x^2+1}{x+1}$ if the curve passes through the point (0,1).

$$\frac{dy}{dx} = \frac{x^2+1}{x+1}$$

$$y = \int \frac{x^2+1}{x+1} dx$$

$$y = \int \left(x-1 + \frac{2}{x+1} \right) dx$$

$$y = \frac{x^2}{2} - x + 2 \ln|x+1| + C$$

Substitute (0,1)

$$1 = \frac{0^2}{2} - 0 + 2 \ln|0+1| + C$$

$$1 = C$$

$\therefore$ Equation of the curve is

$$y = \frac{x^2}{2} - x + 2 \ln|x+1| + 1,$$

20. Find the following indefinite integrals:

(a) $\int \frac{4}{x} dx$

(b) $\int \frac{4}{x^5} dx$

(c) $\int \frac{dx}{2x-8}$

(d) $\int \frac{dx}{(2x-8)^2}$

(e) $\int \frac{x}{3x^2+7} dx$

(f) $\int \frac{x}{(3x^2+7)^3} dx$

(a) $\int \frac{4}{x} dx = 4 \ln|x| + C$

(b) $\int \frac{4}{x^5} dx = 4 \int x^{-5} dx$
 $= 4 \left(\frac{x^{-4}}{-4} \right) + C$
 $= -\frac{1}{x^4} + C$

(e) $\int \frac{6x}{3x^2+7} dx$
 $= \frac{1}{6} \ln|3x^2+7| + C$

(f) $\int \frac{x}{(3x^2+7)^3} dx = \frac{1}{6} \int 6x (3x^2+7)^{-3} dx$
 $= \frac{1}{6} \frac{(3x^2+7)^{-2}}{-2} + C$

(c) $\int \frac{dx}{2x-8} = \frac{1}{2} \ln|2x-8| + C$

$$= \frac{1}{-12(3x^2+7)^2} + C$$

(d) $\int (2x-8)^{-2} dx = \frac{(2x-8)^{-1}}{(-1) \cdot 2} + C$
 $= -\frac{1}{2(2x-8)} + C$

Indefinite Integrals Involving Logarithmic Functions continued

21. Evaluate each of the following definite integrals:

(a) $\int_1^e \frac{(\ln x)^2}{x} dx$

(b) $\int_0^{\frac{\pi}{3}} \tan x dx$

(c) $\int_1^e \frac{(1 + \ln x)^2}{2x} dx$

(d) $\int_e^{e^2} \frac{dw}{w \ln w}$

(e) $\int_1^3 \frac{y dy}{\sqrt{y^2 + 7}}$

(f) $\int_0^2 \frac{x}{(x^2 + 1)^3} dx$

(a) $\int_1^e \frac{1}{x} (\ln x)^2 dx = \left[\frac{(\ln x)^3}{3} \right]_1^e$
 $= \frac{1}{3} [1 - 0] = \frac{1}{3}$

(c) $\frac{1}{2} \int_1^e \frac{1}{x} \cdot (1 + \ln x)^2 dx$
 $= \frac{1}{2} \left[\frac{(1 + \ln x)^3}{3} \right]_1^e$
 $= \frac{1}{6} [8 - 1] = \frac{7}{6}$

(e) $\frac{1}{2} \int_1^3 2y \cdot (y^2 + 7)^{-\frac{1}{2}} dy$
 $= \frac{1}{2} \left[\frac{(y^2 + 7)^{\frac{1}{2}}}{\frac{1}{2}} \right]_1^3 = \sqrt{16} - \sqrt{8}$
 $= 4 - 2\sqrt{2}$

(b) $\int_0^{\frac{\pi}{3}} \tan x dx = - \int_0^{\frac{\pi}{3}} \frac{\sin x}{\cos x} dx$
 $= - \left[\ln |\cos x| \right]_0^{\frac{\pi}{3}}$
 $= - \left[\ln \frac{1}{2} - \ln 1 \right]$
 $= - \ln \left(\frac{1}{2} \right) = \ln 2$

(d) $\int_e^{e^2} \frac{1}{w \ln w} dw$
 $= \left[\ln |\ln w| \right]_e^{e^2}$
 $= \ln 2 - 0 = \ln 2$

(f) $\frac{1}{2} \int_0^2 2x (x^2 + 1)^{-3} dx$
 $= \frac{1}{2} \left[\frac{(x^2 + 1)^{-2}}{-2} \right]_0^2$
 $= \frac{1}{4} \left[\frac{1}{25} - 1 \right]$
 $= \frac{-1}{4} \left[\frac{-24}{25} \right] = \frac{6}{25}$

Applications of Calculus Involving Logarithmic Functions

22. Find the equation of the tangent and the normal to the curve $y = \frac{\ln x}{x}$ at the point where it crosses the x -axis. $\rightarrow y=0$

$y = \frac{\ln x}{x}$

$\frac{\ln x}{x} = 0 \Rightarrow \ln x = 0$
 $\Rightarrow x = e^0$
 $x = 1 \rightarrow (1, 0)$

$y' = \frac{x \cdot \frac{1}{x} - \ln x \cdot 1}{x^2}$

$y' = \frac{1 - \ln x}{x^2}$

at $x=1$, $m_T = \frac{1 - \ln(1)}{1^2} = \frac{1 - 0}{1} = 1$

$m_N = \frac{-1}{m_T} = \frac{-1}{1} = -1$

Equation of the tangent at $(1, 0)$:

$y - 0 = 1(x - 1)$

$y = x - 1$

Equation of the normal:

$y - 0 = -1(x - 1)$

$y = -x + 1$

Applications of Calculus Involving Logarithmic Functions continued

23. The total cost \$C\$ of producing \$x\$ textbooks is given by the function
 $C(x) = 800 + 100x - 100 \ln x$ for $x \geq 1$.

- (a) Find the function $A(x)$, an expression in x for the average cost \$A\$ of producing 1 textbook.
- (b) Find the minimum average cost per book.

$$\textcircled{a} \quad A(x) = \frac{800 + 100x - 100 \ln x}{x}$$

$$A(x) = 100 + \frac{800}{x} - \frac{100 \ln x}{x}, \quad x \geq 1$$

$$\textcircled{b} \quad A'(x) = 0 - \frac{800}{x^2} - 100 \left(\frac{x^{-\frac{1}{x}} - \ln x \cdot x^{-1}}{x^2} \right)$$

$$A'(x) = -\frac{800}{x^2} - 100 \left(\frac{1 - \ln x}{x^2} \right)$$

S.P.:

$$A'(x) = 0$$

$$-\frac{800}{x^2} = \frac{100(1 - \ln x)}{x^2}$$

$$-8 = 1 - \ln x$$

$$-8 - 1 = -\ln x$$

$$\ln x = 9 \Rightarrow x = e^9$$

$$A'(x) = -800 \cdot x^{-2} - 100 \left(\frac{1 - \ln x}{x^2} \right)$$

$$A''(x) = -800(-2 \cdot x^{-3}) - 100 \left[\frac{x^2 \left(-\frac{1}{x^2} \right) - (1 - \ln x) \cdot 2x}{x^4} \right]$$

$$= \frac{1600}{x^3} - 100 \left[\frac{-x - 2x + 2x \ln x}{x^4} \right]$$

$$= \frac{1600}{x^3} - 100 \left(\frac{x(2 \ln x - 3)}{x^4} \right)$$

$$= \frac{1600 - 200 \ln x + 300}{x^3}$$

$$A''(x) = \frac{1900 - 200 \ln x}{x^3}$$

$$\text{at } x = e^9, \quad A''(e^9) = \frac{1900 - 200 \ln(e^9)}{(e^9)^3} = \frac{100}{e^{27}} > 0$$

Minimum average cost at $x = e^9$

$$\therefore x = e^9$$

$$A(e^9) = 100 + \frac{800}{e^9} - \frac{100 \cdot \ln(e^9)}{e^9}$$

$$= 100 - \frac{100}{e^9}$$

$$\approx \$99.99$$

Applications of Calculus Involving Logarithmic Functions continued

24. Consider the function $y = \frac{\ln x}{x}$.

- (a) Find the domain. $x > 0$
- (b) Find the x -intercept. $y = 0 \Rightarrow \frac{\ln x}{x} = 0 \Rightarrow \ln x = 0 \Rightarrow x = e^0 = 1$
- (c) Find the values of x for which the graph of the function is below the x -axis.
- (d) Find the coordinates of the stationary point and determine its nature.
- (e) Find the equation of the vertical asymptote of the function $y = \frac{\ln x}{x}$ and,

given that $\lim_{x \rightarrow \infty} \frac{\ln x}{x} = 0$, write down the equation of the horizontal asymptote.

- (f) Show that $\left(e^{\frac{3}{2}}, \frac{3e^{-\frac{3}{2}}}{2}\right)$ is a point of inflection.

- (g) Sketch the graph of the function $y = \frac{\ln x}{x}$.

(c) $y < 0$

$$\frac{\ln x}{x} < 0$$

$$\Rightarrow 0 < x < 1$$

(d) $y = \frac{\ln x}{x}$

$$y' = \frac{x \cdot \frac{1}{x} - \ln x \cdot 1}{x^2}$$

$$y' = \frac{1 - \ln x}{x^2}$$

S.P.: $y' = 0$

$$\frac{1 - \ln x}{x^2} = 0$$

$$1 - \ln x = 0$$

$$1 = \ln x$$

$$\Rightarrow x = e^1 = e, y = \frac{\ln e}{e} = \frac{1}{e}$$

$\therefore (e, \frac{1}{e})$ is the stationary point

$$y' = \frac{1 - \ln x}{x^2}$$

$$y'' = \frac{x^2(0 - \frac{1}{x}) - (1 - \ln x) \cdot 2x}{x^4}$$

$$= \frac{-x - 2x + 2x \ln x}{x^4}$$

$$= \frac{x(2 \ln x - 3)}{x^4} = \frac{2 \ln x - 3}{x^3}$$

$$\therefore y'(x) = \frac{2 \ln x - 3}{x^3}$$

$$x = e, y'(e) = \frac{2 \ln e - 3}{e^3} = \frac{-1}{e^3} < 0$$

Maximum T.P at $(e, \frac{1}{e})$

(e) V.A: $x = 0$ ($y \rightarrow \pm \infty$)
H.A: $y = 0$ ($x \rightarrow \pm \infty$)

(f) $y''(x) = 0$

$$\frac{2 \ln x - 3}{x^3} = 0$$

$$2 \ln x = 3$$

$$\ln x = \frac{3}{2}$$

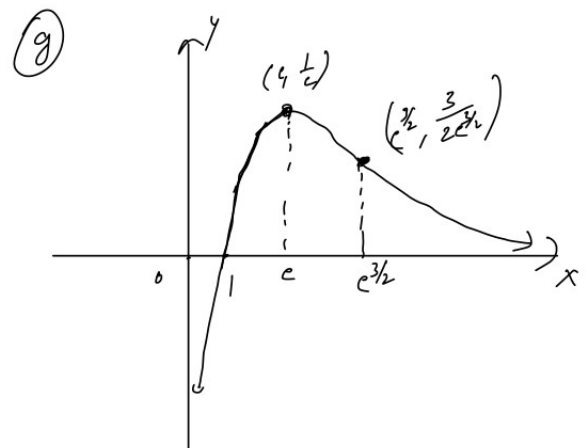
$$x = e^{\frac{3}{2}}, y = \frac{\ln e^{\frac{3}{2}}}{e^{\frac{3}{2}}} = \frac{3}{2e^{\frac{3}{2}}}$$

$(e^{\frac{3}{2}}, \frac{3}{2e^{\frac{3}{2}}})$ is the possible P.O.I

x	1	$e^{\frac{3}{2}}$	∞
y''	-	0	+

concavity changes

$\therefore (e^{\frac{3}{2}}, \frac{3}{2e^{\frac{3}{2}}})$ is the P.O.I



Applications of Calculus Involving Logarithmic Functions continued

25. Find the area of the region bounded by the line $x + y = 5$ and the curve $xy = 4$.

$$y = 5 - x \quad \begin{matrix} 5 - x = \frac{4}{x} \\ y = \frac{4}{x} \end{matrix}$$

$$5x - x^2 = 4$$

$$-x^2 + 5x - 4 = 0$$

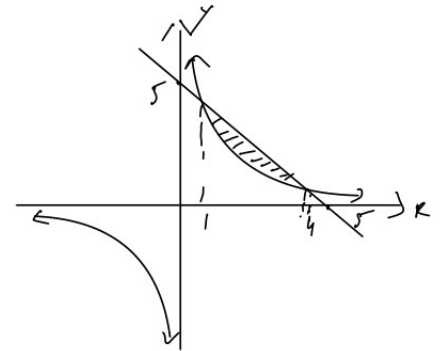
$$(x-1)(x-4) = 0$$

$$x = 1 \text{ or } x = 4$$

$$y = 4 \quad y = 1$$

$$(1, 4) \quad (4, 1)$$

$$\begin{aligned} \text{Area, } A &= \int_1^4 \left(5 - x - \frac{4}{x} \right) dx \\ &= \left[5x - \frac{x^2}{2} - 4 \ln|x| \right]_1^4 \\ &= \left(20 - 8 - 4 \ln 4 \right) - \left(5 - \frac{1}{2} - 4 \ln 1 \right) \\ &= 12 - 4 \ln 4 - 4 \frac{1}{2} \\ A &= \left(7 \frac{1}{2} - 4 \ln 4 \right) \text{ units}^2 \quad \checkmark \end{aligned}$$



26. The region bounded by the curve $y = \sqrt{\frac{x}{x^2 + 1}}$, the x -axis and the line $x = 2$ is rotated about the x -axis. Find the volume of the solid of revolution.
[Note that the curve passes through the origin and lies in the first quadrant.]

$$\begin{aligned} V &= \pi \int_0^2 \left(\sqrt{\frac{x}{x^2 + 1}} \right)^2 dx \\ &= \frac{\pi}{2} \int_0^2 \frac{2x}{x^2 + 1} dx \\ &= \frac{\pi}{2} \left[\ln(x^2 + 1) \right]_0^2 \\ &= \frac{\pi}{2} [\ln 5 - \ln 1] \\ &= \frac{\pi}{2} \ln 5 \text{ units}^3 \end{aligned}$$

Differentiation of Trigonometric Functions

27. Differentiate with respect to x :

(g) $\tan(4x^2 - 1)$

(h) $\cos x^\circ$

(i) $x + 3 \sin x - \cos 2x$

(j) $4 \sin(3x - 5)$

(k) $x^3 \cos 2x$

(l) $\sin 2x \tan 3x$

(m) $\frac{4x}{\cos x}$

(n) $\frac{x+1}{\sin x}$

(o) $(\sin x - \cos x)^2$

(g) $[\tan(4x^2 - 1)]' = 8x \cdot \sec^2(4x^2 - 1)$

(h) $[\cos x^\circ]' = \left[\cos \frac{\pi x}{180}\right]' = -\frac{\pi}{180} \sin x^\circ$

(i) $[x + 3 \sin x - \cos 2x]' = 1 + 3 \cos x + 2 \sin 2x$

(j) $[4 \sin(3x - 5)]' = 12 \cos(3x - 5)$

(k) $[x^3 \cos 2x]' = -2x^3 \sin 2x + 3x^2 \cos 2x = x^2(3 \cos 2x - 2x \sin 2x)$

(l) $[\sin 2x \cdot \tan 3x]' = \sin 2x(3 \sec^2 3x) + \tan 3x \cdot (2 \cos 2x)$

(m) $\left[\frac{4x}{\cos x}\right]' = \frac{\cos x \cdot 4 - 4x(-\sin x)}{\cos^2 x} = \frac{4 \cos x + 4x \sin x}{\cos^2 x}$

(n) $\left[\frac{x+1}{\sin x}\right]' = \frac{\sin x \cdot (1) - (x+1) \cdot \cos x}{\sin^2 x} = \frac{\sin x - (x+1) \cos x}{\sin^2 x}$

(o) $[(\sin x - \cos x)^2]' = (\sin^2 x - 2 \sin x \cos x + \cos^2 x)'$
 $= (1 - \sin 2x)'$
 $= 0 - 2 \cos 2x$
 $= -2 \cos 2x$

Differentiation of Trigonometric Functions continued

28. Find the second derivative of:

(a) $\cos x + \sin x$

$$\begin{aligned} y &= \cos x + \sin x \\ y' &= -\sin x + \cos x \\ y'' &= -\cos x - \sin x \\ y'' &= -(\cos x + \sin x) \end{aligned}$$

(b) $\cos^2 x$

$$\begin{aligned} y &= \cos^2 x \\ y' &= 2\cos x (-\sin x) \\ y' &= -\sin 2x \\ y'' &= -\cos 2x \cdot 2 \\ y'' &= -2\cos 2x \end{aligned}$$

(c) $e^{\sin x}$

$$\begin{aligned} y &= e^{\sin x} \\ y' &= e^{\sin x} \cdot \cos x \\ y' &= e^{\sin x} (-\sin x) + \cos x \cdot e^{\sin x} \cdot \cos x \\ y'' &= e^{\sin x} (-\sin x + \cos^2 x) \end{aligned}$$

29. Find the equation of the normal to the curve $y = \frac{1}{\pi} \cos\left(\pi x - \frac{\pi}{3}\right)$ at the point where $x = 1$. Give your answer in general form.

$$\begin{aligned} y &= \frac{1}{\pi} \cos\left(\pi x - \frac{\pi}{3}\right) \\ y' &= \frac{1}{\pi} \left[-\sin\left(\pi x - \frac{\pi}{3}\right)\right] \cdot \pi \end{aligned}$$

$$y' = -\sin\left(\pi x - \frac{\pi}{3}\right)$$

$$\begin{aligned} \text{at } x=1, \quad m_T &= -\sin\left(\pi - \frac{\pi}{3}\right) \\ &= -\sin \frac{\pi}{3} \\ &= -\frac{\sqrt{3}}{2} \end{aligned}$$

$$m_N = \frac{-1}{m_T} = \frac{-1}{-\frac{\sqrt{3}}{2}} = \frac{2}{\sqrt{3}}$$

Point: $x=1$

$$\begin{aligned} y &= \frac{1}{\pi} \cos\left(\pi - \frac{\pi}{3}\right) \\ &= \frac{1}{\pi} (-\cos \frac{\pi}{3}) \\ &= \frac{1}{\pi} \left(-\frac{1}{2}\right) = -\frac{1}{2\pi} \end{aligned}$$

$(1, -\frac{1}{2\pi})$

Equation of Normal:

$$y + \frac{1}{2\pi} = \frac{2}{\sqrt{3}}(x-1)$$

$$y = \frac{2}{\sqrt{3}}x - \frac{2}{\sqrt{3}} - \frac{1}{2\pi}$$

$$y = \frac{4\pi x - 4\pi - \sqrt{3}}{2\sqrt{3}\pi}$$

$$4\pi x - 2\sqrt{3}\pi y - (4\pi + \sqrt{3}) = 0$$

Differentiation of Trigonometric Functions continued

30. Find the turning point on the curve $y = e^{-x} \sin x$ for $0 \leq x \leq 2$ and determine its nature.

$$y' = e^{-x} (\cos x + \sin x (-e^{-x}))$$

S.P. $y' = 0$

$$e^{-x} (\cos x - \sin x) = 0$$

$$\cos x = \sin x$$

$$1 = \tan x$$

$$x = \frac{\pi}{4} \checkmark (0 \leq x \leq 2)$$

$$x = \frac{\pi}{4}, y = e^{-\frac{\pi}{4}} \cdot \left(\frac{1}{\sqrt{2}}\right)$$

$$\left(\frac{\pi}{4}, \frac{e^{-\frac{\pi}{4}}}{\sqrt{2}}\right) \text{ is the T.P.}$$

Nature:

$$y' = e^{-x} (\cos x - \sin x)$$

$$y'' = e^{-x} (-\sin x - \cos x) + (\cos x - \sin x) \cdot (-e^{-x})$$

$$= e^{-x} (-\sin x - \cos x - \cos x + \sin x)$$

$$y'' = -2 \cos x \cdot e^{-x}$$

at $x = \frac{\pi}{4}, y'' = -2 \cos \frac{\pi}{4} \cdot e^{-\frac{\pi}{4}} < 0$

$\therefore y'' < 0$

Maximum T.P. at $x = \frac{\pi}{4} \checkmark$

31. Consider the function $f(x) = x + \sin x$ for $0 \leq x \leq 2\pi$.

- (a) Find $f(0)$ and $f(2\pi)$.
- (b) Show that there is a horizontal point of inflection at the point where $x = \pi$.
- (c) Except where $x = \pi$, show that $f'(x) > 0$ for all values of x in the domain.
- (d) Sketch the graph of $y = x + \sin x$ for $0 \leq x \leq 2\pi$ showing the above features.

(a) $f(0) = 0 + \sin 0 = 0$

$$f(2\pi) = 2\pi + \sin 2\pi = 2\pi + 0 = 2\pi$$

(b) $f(x) = x + \sin x$ for $0 \leq x \leq 2\pi$

$$f'(x) = 1 + \cos x$$

$$f'(x) = 0$$

$$\cos x = -1$$

$$x = \pi$$

$$f(\pi) = \pi + \sin \pi = \pi + 0 = \pi$$

(π, π) is the turning point

$$f'(x) = 1 + \cos x$$

$$f''(x) = 0 - \sin x = -\sin x$$

$$x = \pi, f''(\pi) = -\sin \pi = 0$$

Second derivative test fails

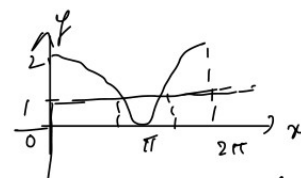
$$f'(x) = 1 + \cos x$$

x	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$
$f'(x)$	+	0	+

H.P.I. at (π, π)

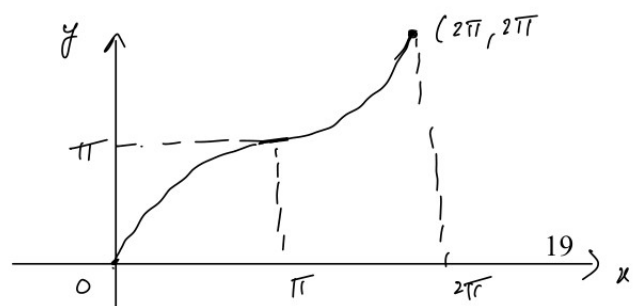
(c) $f'(x) = 1 + \cos x$ ($0 \leq x \leq 2\pi$)

1 + cos x



except $x = \pi, f'(x) > 0$ for all x in the domain

(d)



Integration of Basic Trigonometric Functions

32. Find the primitive (indefinite integral) of the following:

(g) $\sin 2x + \cos 3x$

(h) $\cos(2x-3)$

(i) $\sec^2\left(\frac{1}{3}x-2\right)$

(j) $3\sin\left(\frac{\pi}{2}-x\right)$

(k) $\frac{1}{\cos^2 x}$

(l) $\cos\frac{x}{2} + \frac{1}{2}\sin 2x$

(g) $\int (\sin 2x + \cos 3x) dx = -\frac{\cos 2x}{2} + \frac{\sin 3x}{3} + C$

(h) $\int \cos(2x-3) dx = \frac{\sin(2x-3)}{2} + C$

(i) $\int \sec^2\left(\frac{1}{3}x-2\right) dx = 3 \tan\left(\frac{1}{3}x-2\right) + C$

(j) $\int 3 \sin\left(\frac{\pi}{2}-x\right) dx = \int 3 \cos x dx = 3 \sin x + C$

(k) $\int \frac{1}{\cos^2 x} dx = \int \sec^2 x dx = \tan x + C$

(l) $\int \left(\cos\frac{x}{2} + \frac{1}{2} \sin 2x \right) dx = \frac{\sin\frac{x}{2}}{\frac{1}{2}} + \frac{1}{2} \left(\frac{-\cos 2x}{2} \right) + C$
 $= 2 \sin\frac{x}{2} - \frac{1}{4} \cos 2x + C$

Integration of Basic Trigonometric Functions continued

33. Evaluate the definite integrals:

(g) $\int_0^{\frac{\pi}{4}} (\cos x - \sin x) dx$ (h) $\int_{\frac{\pi}{3}}^{\pi} 1 + \sin 2x dx$ (i) $\int_{-\frac{\pi}{2}}^{\frac{3\pi}{2}} \left(3 \sin 2x - \cos \frac{x}{2} \right) dx$

(j) $\int_0^{\frac{\pi}{3}} \frac{1 + \cos^3 x}{\cos^2 x} dx$ (k) $\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\sec^2 x}{\tan^3 x} dx$ (l) $\int_0^{\frac{\pi}{2}} \sqrt{\sin x} \cos x dx$

(g) $\int_0^{\frac{\pi}{4}} (\cos x - \sin x) dx = [\sin x + \cos x]_0^{\frac{\pi}{4}} = \left(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} \right) - (0 + 1) = \sqrt{2} - 1$

(h) $\int_{\frac{\pi}{3}}^{\pi} (1 + \sin 2x) dx = \left[x - \frac{\cos 2x}{2} \right]_{\frac{\pi}{3}}^{\pi} = \left(\pi - \frac{1}{2} \right) - \left(\frac{\pi}{3} - \frac{1}{2} \left(-\frac{1}{2} \right) \right)$
 $= \frac{2\pi}{3} - \frac{1}{2} - \frac{1}{4} = \frac{2\pi}{3} - \frac{3}{4}$

(i) $\int_{-\frac{\pi}{2}}^{\frac{3\pi}{2}} \left(3 \sin 2x - \cos \frac{x}{2} \right) dx = \left[-\frac{3 \cos 2x}{2} - 2 \sin \frac{x}{2} \right]_{-\frac{\pi}{2}}^{\frac{3\pi}{2}} = \left[-\frac{3}{2} (1) - 2 \left(\frac{1}{2} \right) \right] - \left[-\frac{3}{2} (-1) - 2 \left(-\frac{\sqrt{2}}{2} \right) \right]$
 $= -\frac{3}{2} - \sqrt{2} - \frac{3}{2} - \sqrt{2} = -2\sqrt{2}$

34. Given that $\frac{d^2 y}{dx^2} = -12 \cos 2x - 16 \sin 2x$ and when $x=0$, $\frac{dy}{dx} = 8$ and $y=8$, find y in terms of x .

$$\frac{d^2 y}{dx^2} = -12 \cos 2x - 16 \sin 2x$$

Integrate term on both sides

$$\frac{dy}{dx} = -12 \frac{\sin 2x}{2} + 16 \frac{\cos 2x}{2} + C_1$$

$$\frac{dy}{dx} = -6 \sin 2x + 8 \cos 2x + C_1$$

when $x=0$, $\frac{dy}{dx} = 8$

$$8 = -6(0) + 8(1) + C_1$$

$$\Rightarrow C_1 = 0$$

$$\therefore \frac{dy}{dx} = -6 \sin 2x + 8 \cos 2x$$

Integrate again on both sides
 $y = -6 \left(-\frac{\cos 2x}{2} \right) + 8 \left(\frac{\sin 2x}{2} \right) + C_2$

$$y = 3 \cos 2x + 4 \sin 2x + C_2$$

when $x=0$, $y=8$

$$8 = 3(1) + 4(0) + C_2$$

$$C_2 = 5$$

$$\therefore y = 3 \cos 2x + 4 \sin 2x + 5$$

33

$$\begin{aligned}
 (j) \quad \int_0^{\pi/3} \frac{1 + \cos^3 x}{\cos^2 x} dx &= \int_0^{\pi/3} \left(\frac{1}{\cos^2 x} + \frac{\cos^3 x}{\cos^2 x} \right) dx = \int_0^{\pi/3} (\sec^2 x + \cos x) dx \\
 &= \left[\tan x + \sin x \right]_0^{\pi/3} = \left(\sqrt{3} + \frac{1}{2} \right) - (0) \\
 &= \frac{2\sqrt{3}}{2}
 \end{aligned}$$

$$\begin{aligned}
 (k) \quad \int_{\pi/4}^{\pi/3} \frac{\sec^2 x}{\tan^3 x} dx &= \int_{\pi/4}^{\pi/3} \sec^2 x \cdot (\tan x)^{-3} dx = \left[\frac{(\tan x)^{-2}}{-2} \right]_{\pi/4}^{\pi/3} \\
 &= \frac{1}{-2} \left[\frac{1}{\tan^2 x} \right]_{\pi/4}^{\pi/3} \\
 &= \frac{1}{-2} \left[\frac{1}{3} - \frac{1}{1} \right] \\
 &= \frac{1}{-2} \left[\frac{-2}{3} \right] = \frac{1}{3}
 \end{aligned}$$

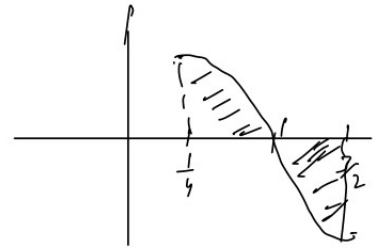
$$\begin{aligned}
 (l) \quad \int_0^{\pi/2} \sqrt{\sin x} \cos x dx &= \int_0^{\pi/2} (\cos x) (\sin x)^{1/2} dx = \left[\frac{(\sin x)^{3/2}}{3/2} \right]_0^{\pi/2} \\
 &= \frac{2}{3} [1 - 0] = \frac{2}{3} \checkmark
 \end{aligned}$$

Integration of Basic Trigonometric Functions continued

35. Find the area bounded by the curve $y = \sin \pi x$ and the x -axis, between $x = \frac{1}{4}$ and

$$x = \frac{3}{2}.$$

$$\begin{aligned} \text{Area } A &= \int_{\frac{1}{4}}^1 \sin \pi x \, dx + \left| \int_1^{\frac{3}{2}} \sin \pi x \, dx \right| \\ &= -\left[\frac{\cos \pi x}{\pi} \right]_{\frac{1}{4}}^1 + \left| -\left[\frac{\cos \pi x}{\pi} \right]_1^{\frac{3}{2}} \right| \\ &= \frac{-1}{\pi} \left(-1 - \frac{1}{\sqrt{2}} \right) + \left| \frac{-1}{\pi} [0 - (-1)] \right| \\ &= \frac{1}{\pi} + \frac{\sqrt{2}}{2\pi} + \frac{1}{\pi} = \frac{4 + \sqrt{2}}{2\pi} \text{ units}^2 \end{aligned}$$



36. The region bounded by the curve $y = \tan 2x$ and the x -axis, from $x = 0$ to $x = \frac{\pi}{6}$ is rotated about the x -axis to form a solid.

(a) Show that the volume of the solid is given by $V = \pi \int_0^{\frac{\pi}{6}} (\sec^2 2x - 1) dx$.

- (b) Find the exact volume of the solid.

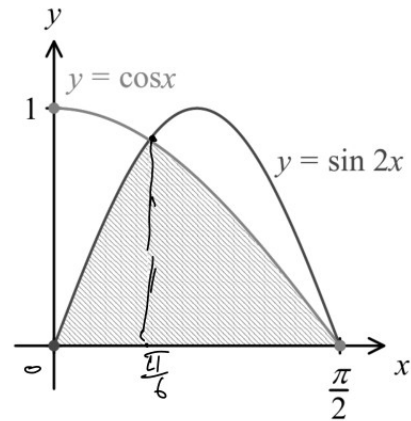
(a) Volume, $V_k = \pi \int_0^{\frac{\pi}{6}} (\tan^2 2x) dx = \pi \int_0^{\frac{\pi}{6}} (\sec^2 2x - 1) dx$

(b)
$$\begin{aligned} V_k &= \pi \left[\frac{\tan 2x}{2} - x \right]_0^{\frac{\pi}{6}} \\ &= \pi \left[\frac{\sqrt{3}}{2} - \frac{\pi}{6} \right] \text{ units}^3 \end{aligned}$$

Integration of Basic Trigonometric Functions continued

37. The diagram shows the graphs of the functions $y = \cos x$ and $y = \sin 2x$ between $x = 0$ and $x = \frac{\pi}{2}$.

Calculate the area of the shaded region.



$$\begin{aligned} \cos x &= \sin 2x \\ \cos x &= 2 \sin x \cos x \\ \cos x (1 - 2 \sin x) &= 0 \\ \cos x &= 0 & 2 \sin x &= 1 \\ x &= \frac{\pi}{2} & \sin x &= \frac{1}{2} \\ & & x &= \frac{\pi}{6} \end{aligned}$$

$$\begin{aligned} \text{Area, } A &= \int_0^{\pi/6} \sin 2x \, dx + \int_{\pi/6}^{\pi/2} \cos x \, dx \\ &= -\left(\frac{\cos 2x}{2}\right)_0^{\pi/6} + \left[\sin x\right]_{\pi/6}^{\pi/2} \\ &= -\frac{1}{2}\left[\frac{1}{2} - 1\right] + \left[1 - \frac{1}{2}\right] \\ &= \frac{1}{4} + \frac{1}{2} \\ &= \frac{3}{4} \text{ units}^2 \end{aligned}$$

Further Calculus of Trigonometric Functions

38. Find the following:

(a) $\int \cos^2 3x \, dx$

(b) $\int 3 \sin^2 \frac{x}{2} \, dx$

(c) $\int 2 - 3 \sin^2 x \, dx$

(d) $\int (1 + \cos x)^2 \, dx$

(e) $\int \sin^2(2x-1) \, dx$

(f) $\int \frac{\cos(2x-1)}{\sin^2(2x-1)} \, dx$

(g) $\int (\cos^4 x - \sin^4 x) \, dx$

(h) $\int 2 \sin x \cos 2x \, dx$

(i) $\int \frac{1 + \sin^2 x}{\cos^2 x} \, dx$

(a) $\int \cos^2 3x \, dx = \int \left(\frac{1 + \cos 6x}{2} \right) dx = \frac{1}{2} x + \frac{\sin 6x}{12} + C$

(b) $\int 3 \sin^2 \frac{x}{2} \, dx = 3 \int \left(\frac{1 - \cos x}{2} \right) dx = \frac{3}{2} \left[x - \sin x \right] + C$

(c) $\int (2 - 3 \sin^2 x) \, dx = \int \left(2 - 3 \left(\frac{1 - \cos 2x}{2} \right) \right) dx = \int \left(2 - \frac{3}{2} + \frac{3}{2} \cos 2x \right) dx = \int \left(\frac{1}{2} + \frac{3}{2} \cos 2x \right) dx = \frac{x}{2} + \frac{3 \sin 2x}{4} + C$

(d) $\int (1 + \cos x)^2 \, dx = \int (1 + 2 \cos x + \cos^2 x) \, dx = \int \left(1 + 2 \cos x + \frac{1 + \cos 2x}{2} \right) dx = \int \left(\frac{3}{2} + 2 \cos x + \frac{1}{2} \cos 2x \right) dx = \frac{3x}{2} + 2 \sin x + \frac{\sin 2x}{4} + C$

(e) $\int \sin^2(2x-1) \, dx = \int \left(\frac{1 - \cos 2(2x-1)}{2} \right) dx = \frac{1}{2} \int (1 - \cos(4x-2)) \, dx = \frac{1}{2} \left[x - \frac{\sin(4x-2)}{4} \right] + C$

(f) $\int \frac{\cos(2x-1)}{\sin^2(2x-1)} \, dx = \frac{1}{2} \int 2 \cos(2x-1) \cdot \left[\sin(2x-1) \right]^{-2} \, dx = \frac{1}{2} \cdot \frac{\left[\sin(2x-1) \right]^{-1}}{-1} + C = \frac{-1}{2 \sin(2x-1)} + C$

(g) $\int (\cos^4 x - \sin^4 x) \, dx = \int (\cos^2 x - \sin^2 x)(\cos^2 x + \sin^2 x) \, dx = \int \cos 2x \, dx = \frac{\sin 2x}{2} + C$

(h) $\int 2 \sin x \cos 2x \, dx = \int 2 \sin x (2 \cos^2 x - 1) \, dx = \int (4 \sin x \cos^2 x - 2 \sin x) \, dx = 4 \int \sin x \cos^2 x \, dx - 2 \int \sin x \, dx = -4 \frac{\cos^3 x}{3} - 2(-\cos x) + C = -\frac{4}{3} \cos^3 x + 2 \cos x + C$

(i) $\int \frac{1 + \sin^2 x}{\cos^2 x} \, dx = \int (\sec^2 x + \tan^2 x) \, dx = \int \sec^2 x + (\sec^2 x - 1) \, dx = \int (2 \sec^2 x - 1) \, dx = 2 \tan x - x + C$

Further Calculus of Trigonometric Functions continued

39. Evaluate the following:

(a) $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^2 x \, dx$

(b) $\int_0^{\frac{\pi}{2}} \sin^2 x \cos x \, dx$

(c) $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} (\sin x + \cos x)^2 \, dx$

(d) $\int_{\frac{\pi}{3}}^{\frac{5\pi}{6}} \frac{\sec^2 x}{(\tan x - 1)^2} \, dx$

(a) $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^2 x \, dx = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1 + \cos 2x}{2} \, dx = \frac{1}{2} \left[x + \frac{\sin 2x}{2} \right]_{-\frac{\pi}{2}}^{\frac{\pi}{2}} = \frac{1}{2} \left[\left(\frac{\pi}{2} + 0 \right) - \left(-\frac{\pi}{2} + 0 \right) \right] = \frac{\pi}{2}$

(b) $\int_0^{\frac{\pi}{2}} \sin^2 x \cos x \, dx = \int_0^{\frac{\pi}{2}} \cos x \cdot (\sin x)^2 \, dx = \left[\frac{(\sin x)^3}{3} \right]_0^{\frac{\pi}{2}} = \frac{1}{3} - 0 = \frac{1}{3}$

(c) $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} (\sin x + \cos x)^2 \, dx = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} (\sin^2 x + 2 \sin x \cos x + \cos^2 x) \, dx = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} (1 + \sin 2x) \, dx = \left[x - \frac{\cos 2x}{2} \right]_{\frac{\pi}{6}}^{\frac{\pi}{3}} = \left(\frac{\pi}{3} - \left(-\frac{1}{4} \right) \right) - \left(\frac{\pi}{6} - \frac{1}{4} \right) = \frac{\pi}{6} + \frac{1}{2}$

(d) $\int_{\frac{\pi}{3}}^{\frac{5\pi}{6}} \frac{\sec^2 x}{(\tan x - 1)^2} \, dx = \int_{\frac{\pi}{3}}^{\frac{5\pi}{6}} \sec^2 x \cdot (\tan x - 1)^{-2} \, dx = \left[\frac{(\tan x - 1)^{-1}}{-1} \right]_{\frac{\pi}{3}}^{\frac{5\pi}{6}} = - \left[\frac{1}{\tan x - 1} \right]_{\frac{\pi}{3}}^{\frac{5\pi}{6}} = - \left[\frac{1}{\left(-\frac{1}{\sqrt{3}} \right) - 1} - \frac{1}{\sqrt{3} - 1} \right] = - \left[\frac{\sqrt{3}}{1 - \sqrt{3}} - \frac{1}{\sqrt{3} - 1} \right] = \frac{\sqrt{3}}{\sqrt{3} + 1} + \frac{1}{\sqrt{3} - 1} = \frac{3 - \sqrt{3} + \sqrt{3} + 1}{(\sqrt{3} + 1)(\sqrt{3} - 1)} = \frac{4}{2} = 2$

Further Calculus of Trigonometric Functions continued

40. Given the function $y = \sin^2 x$ for $-\pi \leq x \leq \pi$,

- find the coordinates of the stationary points and determine their nature;
- find the coordinates of the points of inflection;
- find any intercepts on the coordinate axes;
- sketch the graph of $y = \sin^2 x$;
- find the area bounded by the curve, the x -axis, and the ordinates $x = 0$, $x = \pi$.

(a) $y = \sin^2 x$

$$y' = 2 \sin x \cos x$$

$$y' = 0$$

$$\sin 2x = 0, \quad -\pi \leq x \leq \pi$$

$$-2\pi \leq 2x \leq 2\pi$$

$$2x = \pm 2\pi, \pm \pi, 0$$

$$x = \pm \pi, \pm \frac{\pi}{2}, 0$$

Stationary points are $(-\pi, 0), (\pi, 0), (-\frac{\pi}{2}, 1), (\frac{\pi}{2}, 1), (0, 0)$

$$y' = \sin 2x$$

$$y'' = 2 \cos 2x$$

$$x = -\pi, y'' = 2 \cos 2(-\pi) = 2 \cos 2\pi = 2(1) > 0 \rightarrow \text{min at } (-\pi, 0)$$

$$x = \pi, y'' = 2 \cos 2\pi = 2 > 0 \rightarrow \text{min at } (\pi, 0)$$

$$x = -\frac{\pi}{2}, y'' = 2 \cos 2(-\frac{\pi}{2}) = 2 \cos(-\pi) = 2 \cos \pi = -2 < 0 \rightarrow \text{max at } (-\frac{\pi}{2}, 1)$$

$$x = \frac{\pi}{2}, y'' = 2 \cos 2(\frac{\pi}{2}) = 2 \cos(\pi) = -2 < 0 \rightarrow \text{max at } (\frac{\pi}{2}, 1)$$

$$x = 0, y'' = 2 \cos 2(0) = 2 > 0 \rightarrow \text{min at } (0, 0)$$

Max at $(\pm \frac{\pi}{2}, 1)$, Min at $(0, 0), (\pm \pi, 0)$

(b) $y'' = 2 \cos 2x, \quad -2\pi \leq 2x \leq 2\pi$

$$2 \cos 2x = 0$$

$$\cos 2x = 0$$

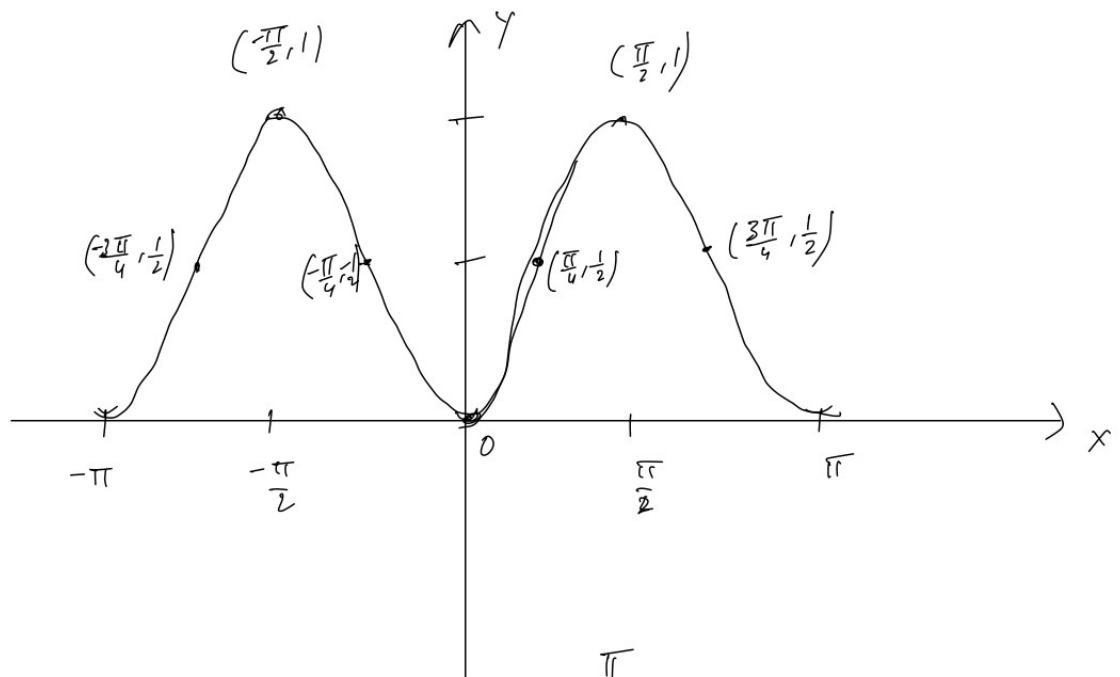
$$2x = \pm \frac{\pi}{2}, \pm \frac{3\pi}{2}$$

$$x = \pm \frac{\pi}{4}, \pm \frac{3\pi}{4}$$

x	$-\pi$	$-\frac{3\pi}{4}$	$-\frac{\pi}{2}$	$-\frac{\pi}{4}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π
y'	+	0	-	0	+	0	-	0	+

concave down

d

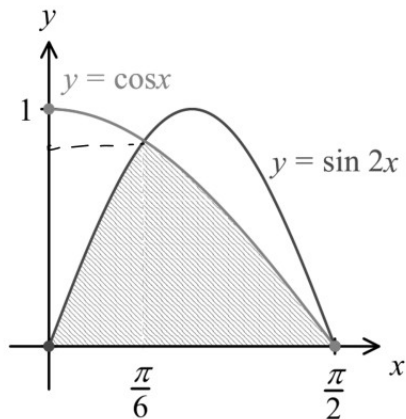


e

$$\begin{aligned}
 \text{Area, } A &= \int_0^{\pi} \sin^2 x \, dx = \int_0^{\pi} \frac{1 - \cos 2x}{2} \, dx \\
 &= \frac{1}{2} \left[x - \frac{\sin 2x}{2} \right]_0^{\pi} \\
 &= \frac{1}{2} \left[(\pi - 0) - (0 - 0) \right] \\
 &= \frac{\pi}{2} \text{ units}^2
 \end{aligned}$$

Further Calculus of Trigonometric Functions continued

41.



The diagram shows the graphs of the functions $y = \cos x$ and $y = \sin 2x$ between $x = 0$ and

$$x = \frac{\pi}{2}.$$

The shaded region is rotated about the x -axis to form a solid of revolution. Find the volume of this solid, giving your answer correct to 3 decimal places.

$$\begin{aligned}
 V_k &= \pi \int_0^{\pi/6} \sin^2 x \, dx + \pi \int_{\pi/6}^{\pi/2} \cos^2 x \, dx \\
 &= \pi \int_0^{\pi/6} \frac{1 - \cos 4x}{2} \, dx + \pi \int_{\pi/6}^{\pi/2} \frac{1 + \cos 2x}{2} \, dx \\
 &= \frac{\pi}{2} \left[x - \frac{\sin 4x}{4} \right]_0^{\pi/6} + \frac{\pi}{2} \left[x + \frac{\sin 2x}{2} \right]_{\pi/6}^{\pi/2} \\
 &= \frac{\pi}{2} \left[\left(\frac{\pi}{6} - \frac{1}{4} \cdot \frac{\sqrt{3}}{2} \right) - (0) \right] + \frac{\pi}{2} \left[\left(\frac{\pi}{2} + 0 \right) - \left(\frac{\pi}{6} + \frac{1}{2} \cdot \frac{\sqrt{3}}{2} \right) \right] \\
 &= \frac{\pi}{2} \left[\frac{\pi}{6} - \frac{\sqrt{3}}{8} \right] + \frac{\pi}{2} \left[\frac{\pi}{3} - \frac{\sqrt{3}}{4} \right] \\
 &= \frac{\pi^2}{12} - \frac{\sqrt{3}\pi}{16} + \frac{\pi^2}{6} - \frac{\sqrt{3}\pi}{8} \\
 &= \frac{\pi^2 + 2\pi^2}{12} - \left(\frac{\sqrt{3}\pi + 2\sqrt{3}\pi}{16} \right) \\
 &= \frac{\pi^2}{4} - \frac{3\sqrt{3}\pi}{16} \\
 &\approx 1.447 \text{ units}^3
 \end{aligned}$$

Rates of Change

42. The number N of bacteria in a culture is given by the formula $N = 2t^4 - t^2 + 2000$, where t is the time in hours.

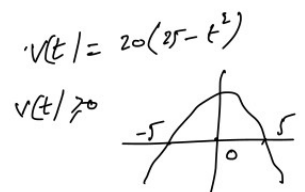
- (a) Find the initial number of bacteria.
 (b) Find the number of bacteria after 5 hours.
 (c) Find the average rate of increase in bacteria in the first 5 hours.
 (d) Find the instantaneous rate of increase in bacteria after 5 hours.

(a) $t=0, N(0) = 2000$
 (b) $t=5, N(5) = 2(5)^4 - 5^2 + 2000 = 3225$ bacteria
 (c) $\text{Average rate} = \frac{N(5) - N(0)}{5 - 0} = \frac{3225 - 2000}{5} = 245$ bacteria/hour
 (d) $N'(t) = 8t^3 - 2t$
 $t=5, N'(5) = 8(5)^3 - 2(5) = 990$ bacteria/hour

43. A 500 litre tank is being emptied of water so that after t minutes, the volume of water in the tank is given by $V = a + bt^2$ where a and b are constants.

- (a) If the tank is initially full of water and takes 5 minutes to empty, find the values of a and b .
 (b) Find the rate at which the volume of water is changing :
 (i) when $t = 2.5$ (ii) when the tank is half empty.

(a) $V(t) = a + bt^2$
 $t=0, V(0) = 500$ litre
 $a + b(0)^2 = 500$
 $a = 500$
 $V(5) = 0$
 $a + b(5)^2 = 0$
 $500 + 25b = 0$
 $25b = -500$
 $b = -20$
 $\therefore V(t) = 500 - 20t^2, 0 \leq t \leq 5$



(b) (i) $V'(t) = -40t$
 $V'(2.5) = -40(2.5) = -100$ L/min

(ii) $V(t) = 250$
 $500 - 20t^2 = 250$
 $250 = 20t^2$
 $t^2 = \frac{25}{2} \Rightarrow t = \frac{5}{\sqrt{2}}$
 $V'\left(\frac{5}{\sqrt{2}}\right) = -40\left(\frac{5}{\sqrt{2}}\right) = -100\sqrt{2}$ L/min

Rates of Change continued

44. The liquid in a vessel flows out through a tap at a rate given by $\frac{dv}{dt} = -10(40-t)$ where $v \text{ cm}^3$ is the volume of liquid in the vessel at time t seconds. Initially the vessel contained 8000 cm^3 of liquid. Find the volume of liquid in the vessel after 10 seconds.

$$\begin{aligned} \frac{dv}{dt} &= -10(40-t) \\ \frac{dv}{dt} &= -400 + 10t \\ \text{Integrate on both sides} \\ v &= -400t + 10 \frac{t^2}{2} + C \\ v &= -400t + 5t^2 + C \\ v(0) &= 8000 \\ 8000 &= -400(0) + 5(0)^2 + C \\ C &= 8000 \end{aligned}$$

$$\begin{aligned} V(t) &= -400t + 5t^2 + 8000 \\ t=10, V(10) &= -400(10) + 5(10)^2 + 8000 \\ &= 4500 \text{ cm}^3 \end{aligned}$$

45. The flow rate of water into a dam over the course of a particular year varies with time and is modelled by $\frac{dW}{dt} = 1.2 - \cos^2 \frac{\pi}{12} t$, where W is the volume of water in the dam measured in thousands of cubic metres and t is the time in months measured from the beginning of January.

- (a) Find the maximum flow rate and the time at which the maximum flow rate occurs.
(b) Given that the dam is initially empty find W as a function of t .
(c) Find the volume of water in the dam at the end of the year.

(a) Let $R(t) = \frac{dW}{dt} = 1.2 - \cos^2 \frac{\pi}{12} t$

$$\begin{aligned} R'(t) &= 0 - 2 \cos \frac{\pi}{12} t \sin \frac{\pi}{12} t \cdot \frac{\pi}{12} \\ R'(t) &= -\sin 2 \left(\frac{\pi}{12} t \right) \cdot \frac{\pi}{12} \\ R'(t) &= -\frac{\pi}{12} \sin \frac{\pi}{6} t \\ R'(t) &= 0 \quad 0 \leq t \leq 12 \\ -\frac{\pi}{12} \sin \frac{\pi}{6} t &= 0 \\ \sin \frac{\pi}{6} t &= 0 \\ \frac{\pi}{6} t &= 0, \pi, 2\pi \\ t &= 0, 6, 12 \\ t &= 0 \text{ or } 6 \text{ or } 12 \end{aligned}$$

$$\begin{aligned} t=0, R(0) &= 1.2 - \cos^2 \frac{\pi}{12} (0) = 1.2 - 1 = 0.2 \\ t=6, R(6) &= 1.2 - \cos^2 \frac{\pi}{12} (6) = 1.2 - 0 = 1.2 \\ t=12, R(12) &= 1.2 - \cos^2 \pi = 1.2 - 1 = 0.2 \end{aligned}$$

$\therefore$ Maximum flow rate is
 $R_{\max} = 1.2 \text{ thousand m}^3/\text{month}$

(b) $\frac{dW}{dt} = 1.2 - \cos^2 \frac{\pi}{12} t$

$$\begin{aligned} \frac{dW}{dt} &= 1.2 - \left(\frac{1 + \cos \frac{\pi}{6} t}{2} \right) \\ &= 1.2 - 0.5 - \frac{1}{2} \cos \frac{\pi}{6} t \\ \frac{dW}{dt} &= 0.7 - 0.5 \cos \frac{\pi}{6} t \\ W &= \int \left(0.7 - 0.5 \cos \frac{\pi}{6} t \right) dt \\ W &= 0.7t - 0.5 \frac{\sin \frac{\pi}{6} t}{\frac{\pi}{6}} + C \\ W &= 0.7t - \frac{3}{\pi} \sin \frac{\pi}{6} t + C \\ t=0, W &= 0 \\ 0 &= 0 - 0 + C \Rightarrow C = 0 \\ \therefore W(t) &= 0.7t - \frac{3}{\pi} \sin \frac{\pi}{6} t \end{aligned}$$

② end of the year, $t=12$

$$w(12) = 0.7(12) - \frac{3}{\pi} \sin \frac{\pi}{6}(12)$$

$$= 8.4 \text{ thousand } m^3$$

$$w(12) = 8400 m^3$$

Exponential Growth and Decay

46. A radioactive substance decomposes from 100g to 90g in 2 hours according to the formula $M = M_0 e^{-kt}$, where t is the number of hours. Find:

- (a) k correct to 3 decimal places,
 (b) the half life of the substance (ie. the time taken for it to decay to half its original mass). Give your answer to the nearest hour.

(a) $M = 100 e^{-kt}$
 $t=2, M(2) = 90$
 $100 e^{-2k} = 90$
 $e^{-2k} = 0.9$
 $-2k = \ln(0.9)$
 $k = \frac{\ln(0.9)}{-2}$
 $k \approx 0.053/h$

(b) $M = 50$
 $100 e^{-kt} = 50$
 $e^{-kt} = 0.5$
 $-kt = \ln(0.5)$
 $t = \frac{\ln(0.5)}{-k}$
 $t = \frac{\ln(0.5)}{-\left(\frac{\ln(0.9)}{-2}\right)}$
 $\therefore t = \frac{2 \ln(0.5)}{\ln(0.9)}$
 $t \approx 13.158$
 $t \approx 13 \text{ hours}$
 (Half life)

47. The rate of change of bacteria in a container is given by $\frac{dN}{dt} = kN$, where N is the population of bacteria in the container and t is the time in hours.

(a) Show that $N = N_0 e^{kt}$, where N_0 is a constant, is a solution to the differential equation $\frac{dN}{dt} = kN$.

(b) If initially there are 2000 bacteria in the container and the bacteria increase at the rate of 2% of the current population, find the number of bacteria in the container after 100 hours.

(a) $N = N_0 e^{kt}$
 $\frac{dN}{dt} = N_0 \cdot e^{kt} \cdot k = kN$
 $\frac{dN}{dt} = kN$

(b) $N_0 = 2000$
 $k = 0.02/h$
 $\therefore N = 2000 e^{0.02t}$
 $t=100, N(100) = 2000 e^{0.02(100)}$
 $= 14778 \text{ bacteria}$

Exponential Growth and Decay continued

48. The number of bacteria N in a colony after t minutes is given by $N = 10\,000 e^{0.05t}$.
Find:

- (i) the number of bacteria after 10 minutes.
- (ii) the time required for the original number to double.
- (iii) the rate at which the colony increases when:
 - (a) $t = 10$.
 - (b) $N = 20\,000$.

(i) $N(t) = 10\,000 e^{0.05t}$
 $t = 10, N(10) = 10\,000 e^{0.05(10)} = 10\,000 e^{0.5} \approx 16\,487 \text{ bacteria}$

(ii) $N = 20\,000$
 $10\,000 e^{0.05t} = 20\,000 \Rightarrow e^{0.05t} = 2 \Rightarrow 0.05t = \ln 2$
 $t = \frac{\ln 2}{0.05} \approx 13.863 \text{ min}$

(iii) $N(t) = 10\,000 e^{0.05t}$
 $N'(t) = 10\,000 e^{0.05t} \cdot 0.05 = 0.05N$
 $N'(t) = 500 e^{0.05t}$
 (a) $t = 10, N'(10) = 500 e^{0.05(10)} \approx 824.361 \text{ bacteria/min}$

(b) $N'(t) = 0.05N$
 when $N = 20\,000, N' = 0.05(20\,000)$
 $= 1000 \text{ bacteria/min}$

Exponential Growth and Decay continued

49. A heated body is cooling and the excess $\theta^\circ\text{C}$ of its temperature above that of its surroundings is given by $\theta = Ae^{-kt}$ where t is the time in minutes.

(i) If at time $t=0$, $\theta=80$, find A .

(ii) If the temperature of the surroundings is 20°C and the body cools to 70°C in 10 minutes, find

(a) its temperature after 20 minutes.

(b) the time taken to reach 60°C .

$$(i) \theta(t) = Ae^{-kt}$$

$$t=0, \theta=80$$

$$80 = A \cdot e^{-k(0)} \Rightarrow A = 80$$

$$\therefore \theta(t) = 80e^{-kt}$$

$$(ii) T = 20 + \theta$$

$$T = 20 + 80e^{-kt}$$

$$\text{given } T(10) = 70$$

$$70 = 20 + 80e^{-10k}$$

$$50 = 80e^{-10k}$$

$$e^{-10k} = \frac{5}{8}$$

$$\Rightarrow -10k = \ln\left(\frac{5}{8}\right)$$

$$k = \frac{-1}{10} \ln\left(\frac{5}{8}\right)$$

$$(ii)$$

$$(b) T = 60$$

$$20 + 80e^{-kt} = 60$$

$$80e^{-kt} = 40$$

$$\Rightarrow e^{-kt} = \frac{1}{2}$$

$$-kt = \ln\left(\frac{1}{2}\right)$$

$$t = \frac{\ln\left(\frac{1}{2}\right)}{-\left(\frac{-1}{10} \ln\left(\frac{5}{8}\right)\right)}$$

$$t = \frac{10 \ln\left(\frac{1}{2}\right)}{\ln\left(\frac{5}{8}\right)}$$

$$t \approx 14.75 \text{ minutes}$$

$$(a) \text{ After 20 minutes } t=20$$

$$T(t) = 20 + 80e^{-kt}$$

$$T(20) = 20 + 80e^{-\left(\frac{-1}{10} \ln\left(\frac{5}{8}\right)\right) \cdot 20}$$

$$= 20 + 80 \cdot \frac{25}{64}$$

$$= 51.25^\circ\text{C}$$

Implicit Differentiation

50. Find $\frac{dy}{dx}$ in each of the following:

(g) $x^3 y^2 = 3$

(h) $xy + y^2 + 6 = 0$

(i) $x^2 y^3 + xy^2 - 8 = 0$

(j) $x^4 + 5x^2 y - 6y^2 x^3 + 3x = 0$

(k) $\frac{x}{y} + 4x = 0$

(l) $x^n y^m = mn$

(g) $x^3 \cdot 2y \cdot y' + y^2 \cdot 3x^2 = 0$
 $y' = \frac{-3x^2 y^2}{2x^3 y} = \frac{-3y}{2x}$

(j) $4x^3 + 5x^2 y' + y \cdot 10x - 6y^2 3x^2 - x^3 12y y' + 3 = 0$
 $y'(5x^2 - 12x^3 y) = -3 - 4x^3 - 10xy + 18x^2 y^2$
 $y' = \frac{18x^2 y^2 - 4x^3 - 10xy - 3}{5x^2 - 12x^3 y}$

(h) $x \cdot y' + y \cdot 1 + 2y y' + 0 = 0$
 $y'(x + 2y) = -y$
 $y' = \frac{-y}{x + 2y}$

(k) $\frac{y \cdot 1 - x y'}{y^2} + 4 = 0$
 $y - x y' = -4y^2$
 $\frac{y + 4y^2}{x} = y'$

(i) $x^2 \cdot 3y^2 y' + y^3 \cdot 2x + x \cdot 2y y' + y^2 \cdot 1 = 0$
 $y'(3x^2 y^2 + 2xy) = -y^2 - 2xy^3$
 $y' = \frac{-y^2(1 + 2x)}{xy(3xy + 2)} = \frac{-y(1 + 2x)}{x(3xy + 2)}$

(l) $x^n \cdot m y^{m-1} y' + y^m \cdot n x^{n-1} = 0$
 $y' = \frac{-n \cdot y^m \cdot x^{n-1}}{m \cdot x^n \cdot y^{m-1}}$

51. If $x = y^2 + y$ find $\frac{dy}{dx}$ and $\frac{d^2 y}{dx^2}$.

$y' = \frac{-n \cdot y \cdot x}{m} \Rightarrow y' = \frac{-n y}{m x}$

$x = y^2 + y$

$1 = 2y \cdot \frac{dy}{dx} + \frac{dy}{dx}$

$\frac{dy}{dx} = \frac{1}{2y + 1}$

$\frac{dy}{dx} = (2y + 1)^{-1}$

$\frac{d^2 y}{dx^2} = (-1)(2y + 1)^{-2} \left(2 \cdot \frac{dy}{dx}\right)$
 $= \frac{-2}{(2y + 1)^2} \cdot \frac{1}{(2y + 1)}$

$\frac{d^2 y}{dx^2} = \frac{-2}{(2y + 1)^3}$

Implicit Differentiation continued

52. For the curve $(x+y)^2 = 4x^2y + 1$ find the equation of the tangent at the point $(1,2)$ and the equation of the normal at the point $(1,0)$.

$$\frac{d}{dx} [(x+y)^2] = \frac{d}{dx} [4x^2y + 1]$$

$$2(x+y) \cdot \left(1 + \frac{dy}{dx}\right) = 4x^2 \cdot \frac{dy}{dx} + y \cdot 8x + 0$$

$$(x+y) + (x+y) \frac{dy}{dx} = 2x^2 \frac{dy}{dx} + 4xy$$

$$\frac{dy}{dx} (x+y-2x^2) = 4xy - x - y$$

$$\frac{dy}{dx} = \frac{4xy - x - y}{x + y - 2x^2}$$

$$\text{at } (1,2), m_T = \frac{8 - 1 - 2}{1 + 2 - 2} = \frac{5}{1} = 5$$

$$\text{at } (1,0), m_N = \frac{-1}{m_T} = \frac{-1}{\frac{5-1}{1-2}} = -1$$

53. Find $\frac{dy}{dx}$ in each of the following:

(a) $e^x + e^y = e^5$

$$\frac{d}{dx} [e^x + e^y] = \frac{d}{dx} (e^5)$$

$$e^x + e^y \frac{dy}{dx} = 0$$

$$e^y \frac{dy}{dx} = -e^x$$

$$\frac{dy}{dx} = \frac{-e^x}{e^y}$$

$$\frac{dy}{dx} = -e^{x-y}$$

(b) $xe^y = y^2 + e^x$

$$\frac{d}{dx} (xe^y) = \frac{d}{dx} (y^2 + e^x)$$

$$x \cdot e^y \frac{dy}{dx} + e^y \cdot 1 = 2y \frac{dy}{dx} + e^x$$

$$\frac{dy}{dx} (xe^y - 2y) = e^x - e^y$$

$$\frac{dy}{dx} = \frac{e^x - e^y}{xe^y - 2y}$$

Equation of the tangent at $(1,2)$:

$$y - 2 = 5(x - 1)$$

$$y = 5x - 3$$

Equation of Normal at $(1,0)$:

$$y - 0 = -1(x - 1)$$

$$y = -x + 1$$

$$y = 1 - x$$

Implicit Differentiation continued

54. Find $\frac{dy}{dx}$ in each of the following :

(a) $x \ln y + y = 20$

$$\frac{d}{dx} [x \ln y + y] = \frac{d}{dx} (20)$$

$$x \cdot \frac{1}{y} \cdot \frac{dy}{dx} + \ln y \cdot 1 + \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} \left(\frac{x}{y} + 1 \right) = -\ln y$$

$$\frac{dy}{dx} \left(\frac{x+y}{y} \right) = -\ln y$$

$$\frac{dy}{dx} = \frac{-y \ln y}{x+y}$$

(b) $\ln x \ln y = 1$

$$\frac{d}{dx} (\ln x \cdot \ln y) = \frac{d}{dx} (1)$$

$$\ln x \cdot \frac{1}{y} \frac{dy}{dx} + \ln y \cdot \frac{1}{x} = 0$$

$$\frac{dy}{dx} = \frac{-\frac{\ln y}{x}}{\frac{\ln x}{y}}$$

$$\frac{dy}{dx} = \frac{-y \ln y}{x \ln x}$$

55. Find $\frac{dy}{dx}$ in each of the following :

(c) $\sin(x-2y) = y$

$$\frac{d}{dx} [\sin(x-2y)] = \frac{d}{dx} (y)$$

$$\cos(x-2y) \cdot \left(1 - 2 \frac{dy}{dx} \right) = \frac{dy}{dx}$$

$$\cos(x-2y) - 2 \cos(x-2y) \frac{dy}{dx} = \frac{dy}{dx}$$

$$\frac{dy}{dx} (1 + 2 \cos(x-2y)) = \cos(x-2y)$$

$$\therefore \frac{dy}{dx} = \frac{\cos(x-2y)}{1 + 2 \cos(x-2y)}$$

(d) $\sin(y+x) = \cos(y-x)$

$$\frac{d}{dx} [\sin(y+x)] = \frac{d}{dx} [\cos(y-x)]$$

$$\cos(y+x) \cdot \left(\frac{dy}{dx} + 1 \right) = -\sin(y-x) \left[\frac{dy}{dx} - 1 \right]$$

$$\frac{dy}{dx} [\cos(y+x) + \sin(y-x)] = \sin(y-x) - \cos(y+x)$$

$$\frac{dy}{dx} = \frac{\sin(y-x) - \cos(y+x)}{\cos(y+x) + \sin(y-x)}$$

Related Rates

56. A spherical balloon is being filled with air at the rate of $100 \text{ cm}^3/\text{min}$. Find the rate at which the radius of the sphere is increasing when the radius is 5 cm .

Given $\frac{dV}{dt} = 100 \text{ cm}^3/\text{min}$

Find $\frac{dr}{dt}$ when $r = 5 \text{ cm}$

Volume of a sphere:

$$V = \frac{4\pi}{3} r^3$$

$$\frac{dV}{dt} = \frac{4\pi}{3} \cdot 3r^2 \cdot \frac{dr}{dt}$$

$$100 = 4\pi \cdot (5)^2 \cdot \frac{dr}{dt}$$

$$\Rightarrow \frac{dr}{dt} = \frac{100}{100\pi} = \frac{1}{\pi} \text{ cm/min}$$

$$\therefore \frac{dr}{dt} = \frac{1}{\pi} \text{ cm/min}$$

57. A ladder 5 metres long has its upper end against a vertical wall and its lower end on the horizontal floor. The lower end moves away from the wall at a constant rate of 1 m/second . Find the rate at which the upper end moves down the wall when the lower end is 4 metres from the wall.

$$x^2 + y^2 = 25$$

$$\frac{d}{dt}(x^2 + y^2) = \frac{d}{dt}(25)$$

$$2x \cdot \frac{dx}{dt} + 2y \cdot \frac{dy}{dt} = 0$$

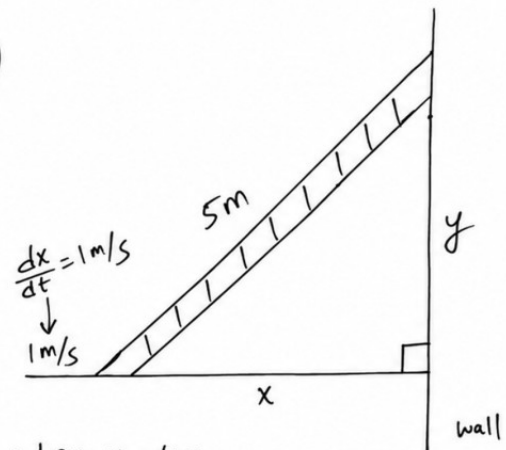
$$y \cdot \frac{dy}{dt} = -x \cdot \frac{dx}{dt}$$

$$\frac{dy}{dt} = -\frac{x}{y} \cdot \frac{dx}{dt}$$

$$= -\frac{4}{3} \cdot (1)$$

$$\frac{dy}{dt} = -\frac{4}{3} \text{ m/s}$$

$\therefore$ upper end moves down at $\frac{4}{3} \text{ m/s}$



when $x = 4 \text{ m}$

$$5^2 = x^2 + y^2$$

$$25 - 16 = y^2$$

$$y = 3$$

Related Rates continued

58. Sand falling forms a conical pile such that the height of the pile is one and a half times the radius.

- (a) Show that the volume of the pile is $V = \frac{4}{27} \pi h^3$.
- (b) If the sand is falling at the rate of $\frac{\pi}{10} \text{ m}^3 / \text{minute}$, find the rate at which the height is increasing when the pile of sand is 3m high.

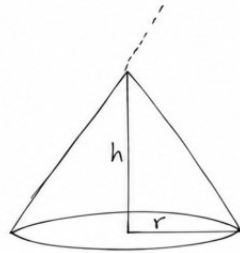
(a)

$$h = 1\frac{1}{2}r$$

$$h = \frac{3r}{2} \Rightarrow r = \frac{2h}{3}$$

Volume of a cone,

$$\begin{aligned} V &= \frac{h}{3} \cdot \pi r^2 \\ &= \frac{h}{3} \pi \left(\frac{2h}{3}\right)^2 \\ &= \frac{h}{3} \cdot \pi \cdot \frac{4h^2}{9} \\ &= \frac{4\pi h^3}{27} \\ V &= \frac{4\pi h^3}{27} \end{aligned}$$



(b)

$$\text{find } \frac{dh}{dt} \text{ when } h = 3\text{m}$$

$$\text{given } \frac{dV}{dt} = \frac{\pi}{10} \text{ m}^3/\text{min}$$

$$V = \frac{4}{27} \pi h^3$$

$$\frac{d}{dt}(V) = \frac{d}{dt}\left(\frac{4\pi}{27} h^3\right)$$

$$\frac{dV}{dt} = \frac{4\pi}{27} \cdot 3h^2 \cdot \frac{dh}{dt}$$

$$\frac{\pi}{10} = \frac{4\pi}{9} \cdot (3)^2 \cdot \frac{dh}{dt}$$

$$\frac{dh}{dt} = \frac{1}{40} \text{ m/min}$$

59. A searchlight on the ground detects and tracks a plane that is due east of the searchlight and at a height of 900 m above the ground. The plane is travelling due West at 240 km per hour. Find the rate of change of the angle of elevation of the plane from the searchlight when the plane is 900 m east of the searchlight.

$$\tan \theta = \frac{900}{x}$$

$$\frac{dx}{dt} = -240 \text{ km/hr}$$

$$= \frac{-240 \times 1000}{3600}$$

$$= -\frac{200}{3} \text{ m/s}$$

Angle increases

as the plane approaches

$$\frac{d}{dt}(\tan \theta) = \frac{d}{dt}\left(\frac{900}{x}\right)$$

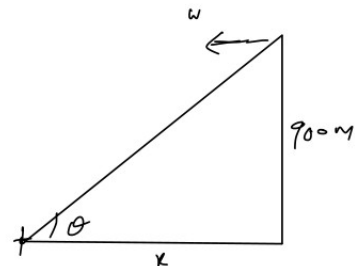
$$\sec^2 \theta \cdot \frac{d\theta}{dt} = 900 \left(-\frac{1}{x^2}\right) \cdot \frac{dx}{dt}$$

$$\text{When } x=900, \theta = \frac{\pi}{4}$$

$$\text{If } \frac{d\theta}{dt} = -\frac{900}{900^2} \cdot \frac{200}{3}$$

$$= -\frac{1}{900} \cdot \frac{200}{3}$$

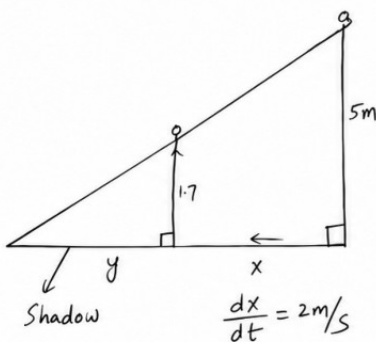
$$= -\frac{1}{27} \text{ rad/sec}$$



Related Rates continued

60. A street light is on top of a pole which is 5 metres high. A person who is 1.7 metres tall walks away from the pole at a constant rate of 2 metres per second. As he walks his shadow lengthens.

- (a) Find the rate at which the length of the person's shadow is increasing.
 (b) Find the rate at which the tip of the person's shadow is moving away from the pole.



By Similar triangles

(a)

$$\frac{5}{x+y} = \frac{1.7}{y}$$

$$5y = 1.7x + 1.7y$$

$$3.3y = 1.7x$$

$$y = \frac{1.7}{3.3}x$$

$$\frac{d}{dt}(y) = \frac{d}{dt}\left(\frac{1.7}{3.3}x\right)$$

$$\frac{dy}{dt} = \frac{1.7}{3.3} \frac{dx}{dt}$$

$$= \frac{1.7}{3.3} \cdot 2$$

$$\frac{dy}{dt} = \frac{3.4}{3.3} \text{ m/s}$$

rate of persons
Shadow

(b)

tip of person's shadow:

$$s = x + y$$

$$\frac{d}{dt}(s) = \frac{d}{dt}(x+y)$$

$$\frac{ds}{dt} = \frac{dx}{dt} + \frac{dy}{dt}$$

$$= 2 + \frac{3.4}{3.3}$$

$$= \frac{100}{33} \text{ m/s.}$$

Introduction to Inverse Trigonometric Functions

61. Without using a calculator write down exact values of :

$$\begin{array}{lll}
 \text{(g)} & \sin^{-1}\left(\frac{1}{\sqrt{2}}\right) & \text{(h)} & \sin^{-1}\left(-\frac{1}{\sqrt{2}}\right) & \text{(i)} & \cos^{-1}\left(\frac{\sqrt{3}}{2}\right) \\
 \text{(j)} & \cos^{-1}\left(-\frac{\sqrt{3}}{2}\right) & \text{(k)} & \tan^{-1}(\sqrt{3}) & \text{(l)} & \tan^{-1}(-\sqrt{3})
 \end{array}$$

$$\begin{array}{ll}
 \text{62} & \begin{array}{l} \sin^{-1}x \\ \cos^{-1}x \end{array} \left\{ \begin{array}{l} -1 \leq x \leq 1 \\ x \in \mathbb{R} \end{array} \right. \\
 \text{(a)} & \sin^{-1}(2) \rightarrow \text{DNE} \\
 \text{(b)} & \cos^{-1}(2) \rightarrow \text{DNE} \\
 \text{(c)} & \tan^{-1}(2) \rightarrow \text{exist} \\
 \text{(d)} & \sin^{-1}(-2) \rightarrow \text{DNE} \\
 \text{(e)} & \tan^{-1}(-2) \rightarrow \text{exist} \\
 \text{(f)} & \sin^{-1}\left(\tan\frac{\pi}{3}\right) = \sin^{-1}(\sqrt{3}) = \text{DNE} \\
 \text{(g)} & \sin^{-1}\left(-\tan\frac{\pi}{6}\right) = \sin^{-1}\left(-\frac{1}{\sqrt{3}}\right) \Rightarrow \text{exist} \\
 \text{61} & \begin{array}{l} \sin^{-1}\left(\frac{1}{\sqrt{2}}\right) = \frac{\pi}{4} \\ \sin^{-1}\left(-\frac{1}{\sqrt{2}}\right) = -\frac{\pi}{4} \\ \cos^{-1}\left(\frac{\sqrt{3}}{2}\right) = \frac{\pi}{6} \\ \cos^{-1}\left(-\frac{\sqrt{3}}{2}\right) = -\frac{\pi}{6} \\ \tan^{-1}(\sqrt{3}) = \frac{\pi}{3} \\ \tan^{-1}(-\sqrt{3}) = -\frac{\pi}{3}
 \end{array}
 \end{array}$$

62. Without using the calculator state which of the following **do not exist**:

$$\begin{array}{llll}
 \text{(a)} & \sin^{-1} 2 & \text{(b)} & \cos^{-1} 2 \\
 \text{(c)} & \tan^{-1} 2 & \text{(d)} & \sin^{-1}(-2) \\
 \text{(e)} & \tan^{-1}(-2) & \text{(f)} & \sin^{-1}\left(\tan\frac{\pi}{3}\right) \\
 \text{(g)} & \sin^{-1}\left(-\tan\frac{\pi}{6}\right)
 \end{array}$$

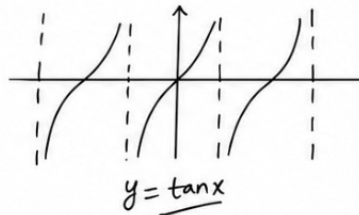
$$\begin{array}{ll}
 \text{62} & \begin{array}{l} \sin^{-1}x \\ \cos^{-1}x \end{array} \left\{ \begin{array}{l} -1 \leq x \leq 1 \\ x \in \mathbb{R} \end{array} \right. \\
 \text{(a)} & \sin^{-1}(2) \rightarrow \text{DNE} \\
 \text{(b)} & \cos^{-1}(2) \rightarrow \text{DNE} \\
 \text{(c)} & \tan^{-1}(2) \rightarrow \text{exist} \\
 \text{(d)} & \sin^{-1}(-2) \rightarrow \text{DNE} \\
 \text{(e)} & \tan^{-1}(-2) \rightarrow \text{exist} \\
 \text{(f)} & \sin^{-1}\left(\tan\frac{\pi}{3}\right) = \sin^{-1}(\sqrt{3}) = \text{DNE} \\
 \text{(g)} & \sin^{-1}\left(-\tan\frac{\pi}{6}\right) = \sin^{-1}\left(-\frac{1}{\sqrt{3}}\right) \Rightarrow \text{exist} \\
 \text{61} & \begin{array}{l} \sin^{-1}\left(\frac{1}{\sqrt{2}}\right) = \frac{\pi}{4} \\ \sin^{-1}\left(-\frac{1}{\sqrt{2}}\right) = -\frac{\pi}{4} \\ \cos^{-1}\left(\frac{\sqrt{3}}{2}\right) = \frac{\pi}{6} \\ \cos^{-1}\left(-\frac{\sqrt{3}}{2}\right) = -\frac{\pi}{6} \\ \tan^{-1}(\sqrt{3}) = \frac{\pi}{3} \\ \tan^{-1}(-\sqrt{3}) = -\frac{\pi}{3}
 \end{array}
 \end{array}$$

Properties of Inverse Trigonometric Functions

63. Consider the function $y = \tan^{-1}(\tan x)$.

- State the domain.
- State the period.
- Show that the function is odd.
- State the values of x for which $\tan^{-1}(\tan x) = x$
- Hence sketch the graph of the function $y = \tan^{-1}(\tan x)$ for $-\frac{3\pi}{2} \leq x \leq \frac{3\pi}{2}$.

3 $y = \tan^{-1}(\tan x)$



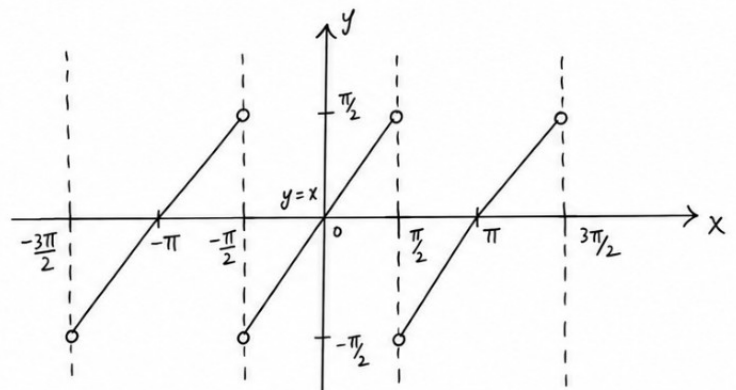
(a) Domain: $x \neq n\pi + \frac{\pi}{2}, n \in \mathbb{Z}$

(b) $f(x) = \tan x$
 $f(x+\pi) = \tan(x+\pi) = \tan x$
 Period is π ✓

(c) $f(x) = \tan^{-1}(\tan x)$
 $f(-x) = \tan^{-1}(\tan(-x)) = \tan^{-1}(-\tan x)$
 $= -\tan^{-1}(\tan x)$
 $= -f(x)$
 $\therefore f(-x) = -f(x)$
 $\therefore f(x)$ is an odd function.

(d) $\tan^{-1}(\tan x) = x$
 when $-\frac{\pi}{2} < x < \frac{\pi}{2}$

(e) $y = \tan^{-1}(\tan x)$ for $-\frac{3\pi}{2} \leq x \leq \frac{3\pi}{2}$



Properties of Inverse Trigonometric Functions continued

64. (a) Prove that $\tan^{-1}\left(\frac{1}{4}\right) + \tan^{-1}\left(\frac{3}{5}\right) = \frac{\pi}{4}$.
- (b) Prove that $\cos^{-1}\left(\frac{3}{\sqrt{10}}\right) + \cos^{-1}\left(\frac{2}{\sqrt{5}}\right) = \frac{\pi}{4}$.
- (c) Prove that $\tan^{-1}\frac{3}{4} = 2 \tan^{-1}\frac{1}{3}$.

64 (a) Prove that $\tan^{-1}\left(\frac{1}{4}\right) + \tan^{-1}\left(\frac{3}{5}\right) = \frac{\pi}{4}$

$$\alpha = \tan^{-1}\left(\frac{1}{4}\right) \quad \beta = \tan^{-1}\left(\frac{3}{5}\right)$$

$$\Leftrightarrow \begin{cases} \tan \alpha = \frac{1}{4} > 0 \\ 0 < \alpha < \frac{\pi}{2} \end{cases} \quad \Leftrightarrow \begin{cases} \tan \beta = \frac{3}{5} > 0 \\ 0 < \beta < \frac{\pi}{2} \end{cases}$$

$$\alpha + \beta = \frac{\pi}{4}$$

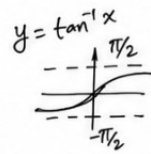
$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \cdot \tan \beta} = \frac{\frac{1}{4} + \frac{3}{5}}{1 - \frac{1}{4} \cdot \frac{3}{5}} = \frac{\frac{5+12}{20}}{\frac{20-3}{20}} = \frac{17}{17} = 1$$

$$\tan(\alpha + \beta) = 1$$

$$\tan(\alpha + \beta) = \tan \frac{\pi}{4}$$

$$\therefore \alpha + \beta = \frac{\pi}{4}$$

hence proved



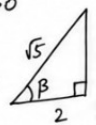
(b) Prove that $\cos^{-1}\left(\frac{3}{\sqrt{10}}\right) + \cos^{-1}\left(\frac{2}{\sqrt{5}}\right) = \frac{\pi}{4}$

$$\alpha = \cos^{-1}\left(\frac{3}{\sqrt{10}}\right) \quad \beta = \cos^{-1}\left(\frac{2}{\sqrt{5}}\right)$$

$$\Leftrightarrow \begin{cases} \cos \alpha = \frac{3}{\sqrt{10}} > 0 \\ 0 < \alpha < \frac{\pi}{2} \end{cases} \quad \Leftrightarrow \begin{cases} \cos \beta = \frac{2}{\sqrt{5}} > 0 \\ 0 < \beta < \frac{\pi}{2} \end{cases}$$



$$\sin \alpha = \frac{1}{\sqrt{10}}$$



$$\sin \beta = \frac{1}{\sqrt{5}}$$

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

$$= \frac{3}{\sqrt{10}} \cdot \frac{2}{\sqrt{5}} - \frac{1}{\sqrt{10}} \cdot \frac{1}{\sqrt{5}}$$

$$= \frac{6-1}{\sqrt{50}} = \frac{5}{5\sqrt{2}} = \frac{1}{\sqrt{2}}$$

$$\cos(\alpha + \beta) = \frac{1}{\sqrt{2}}$$

$$\cos(\alpha + \beta) = \cos \frac{\pi}{4} \Rightarrow \alpha + \beta = \frac{\pi}{4} \quad \text{hence proved}$$

(c) Prove that $\tan^{-1}\frac{3}{4} = 2 \tan^{-1}\frac{1}{3}$

$$\alpha = \tan^{-1}\left(\frac{1}{3}\right)$$

$$\Leftrightarrow \begin{cases} \tan \alpha = \frac{1}{3} \\ 0 < \alpha < \frac{\pi}{2} \end{cases}$$

$$\text{Prove: } \tan^{-1}\frac{3}{4} = 2\alpha$$

$$\Leftrightarrow \tan 2\alpha = \frac{3}{4}$$

$$\tan 2\alpha = \frac{2 \tan \alpha}{1 - \tan^2 \alpha}$$

$$= \frac{2(\frac{1}{3})}{1 - (\frac{1}{3})^2} = \frac{\frac{2}{3}}{\frac{8}{9}}$$

$$= \frac{2}{3} \times \frac{9}{8} = \frac{3}{4}$$

$$\tan 2\alpha = \frac{3}{4}$$

$$\Rightarrow 2\alpha = \tan^{-1}\left(\frac{3}{4}\right)$$

Differentiation of Inverse Trigonometric Functions

65. Assuming all the functions below are differentiable, find the derivatives of :

(a) $\sin^{-1}(x^2)$

(b) $\cos^{-1}\left(\frac{x}{2}\right)$

(c) $\tan^{-1}\left(\frac{3-x}{5}\right)$

(d) $e^x \sin^{-1}(x)$

(e) $\ln[\tan^{-1}(x)]$

(f) $\tan^{-1}\left(\frac{x+2}{1-2x}\right)$

65
(a) $\left(\sin^{-1}(x^2)\right)' = \frac{2x}{\sqrt{1-x^4}}$

(b) $\left(\cos^{-1}\left(\frac{x}{2}\right)\right)' = \frac{-1}{\sqrt{4-x^2}}$

(c) $\left(\tan^{-1}\left(\frac{3-x}{5}\right)\right)' = \frac{-\frac{1}{5}}{1+\left(\frac{3-x}{5}\right)^2}$
 $= \frac{-\frac{1}{5}}{\frac{25+(3-x)^2}{25}} = \frac{-5}{25+(3-x)^2}$

(d) $\left(e^x \cdot \sin^{-1}x\right)'$
 $= e^x \cdot \frac{1}{\sqrt{1-x^2}} + \sin^{-1}x \cdot e^x$
 $= e^x \left(\frac{1}{\sqrt{1-x^2}} + \sin^{-1}x \right)$

(e) $\left(\ln(\tan^{-1}x)\right)'$
 $= \frac{\frac{1}{1+x^2}}{\tan^{-1}x} = \frac{1}{(1+x^2)\tan^{-1}x}$

(f) $\left(\tan^{-1}\left(\frac{x+2}{1-2x}\right)\right)' = \frac{(1-2x) \cdot 1 - (x+2)(-2)}{(1-2x)^2 + \left(\frac{x+2}{1-2x}\right)^2}$
 $= \frac{1-2x+2x+4}{(1-2x)^2 + (x+2)^2}$
 $= \frac{5}{1-4x+4x^2+x^2+4x+4}$
 $= \frac{5}{5x^2+5}$
 $= \frac{1}{x^2+1}$

Differentiation of Inverse Trigonometric Functions continued

66. Find $f''(x)$ for each of the following functions:

(a) $f(x) = \sin^{-1} x$

(b) $f(x) = \sin^{-1} \frac{x}{4}$

(c) $f(x) = \tan^{-1}(x^3)$

① $f(x) = \sin^{-1} x$

$$f'(x) = \frac{1}{\sqrt{1-x^2}}$$

$$f'(x) = (1-x^2)^{-\frac{1}{2}}$$

$$f''(x) = -\frac{1}{2} (1-x^2)^{-\frac{3}{2}} \cdot (-2x)$$

$$f''(x) = \frac{x}{(1-x^2)^{\frac{3}{2}}}$$

② $f(x) = \sin^{-1}\left(\frac{x}{4}\right)$

$$f'(x) = \frac{1}{\sqrt{16-x^2}}$$

$$f'(x) = (16-x^2)^{-\frac{1}{2}}$$

$$f''(x) = -\frac{1}{2} (16-x^2)^{-\frac{3}{2}} \cdot (-2x)$$

$$f''(x) = \frac{x}{(16-x^2)^{\frac{3}{2}}}$$

③ $f(x) = \tan^{-1}(x^3)$

$$f'(x) = \frac{3x^2}{1+(x^3)^2}$$

$$f'(x) = \frac{3x^2}{1+x^6} \left(\frac{u}{v}\right)$$

$$f''(x) = \frac{(1+x^6) \cdot (6x) - 3x^2(6x^5)}{(1+x^6)^2}$$

$$= \frac{6x(1+x^6-3x^6)}{(1+x^6)^2} = \frac{6x(1-2x^6)}{(1+x^6)^2}$$

67. Find $\frac{dy}{dx}$ for each of the following :

(a) $\cos^{-1} x + \sin^{-1} y = \pi$

(b) $\tan^{-1}(2y) = x^2 + y$

(c) $\tan^{-1}(xy) = y$

67
① $\cos^{-1} x + \sin^{-1} y = \pi$

$$\frac{d}{dx}(\sin^{-1} y) = \frac{d}{dx}(\pi - \cos^{-1} x)$$

$$\frac{1}{\sqrt{1-y^2}} \frac{dy}{dx} = 0 - \left(\frac{-1}{\sqrt{1-x^2}}\right)$$

$$\frac{1}{\sqrt{1-y^2}} \frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}}$$

$$\frac{dy}{dx} = \frac{\sqrt{1-y^2}}{\sqrt{1-x^2}}$$

② $\frac{d}{dx}(\tan^{-1}(2y)) = \frac{d}{dx}(x^2 + y)$

$$\frac{2 \frac{dy}{dx}}{1+(2y)^2} = 2x + \frac{dy}{dx}$$

$$\left(\frac{2}{1+4y^2} - 1\right) \frac{dy}{dx} = 2x$$

$$\left(\frac{2-1-4y^2}{1+4y^2}\right) \frac{dy}{dx} = 2x$$

$$\left(\frac{1-4y^2}{1+4y^2}\right) \frac{dy}{dx} = 2x$$

$$\frac{dy}{dx} = \frac{2x(1+4y^2)}{1-4y^2}$$

③ $\frac{d}{dx}(\tan^{-1}(xy)) = \frac{d}{dx}(y)$

$$\frac{x \cdot \frac{dy}{dx} + y \cdot 1}{1+(xy)^2} = \frac{dy}{dx}$$

$$x \cdot \frac{dy}{dx} + y = \frac{dy}{dx} (1+x^2y^2)$$

$$y = \frac{dy}{dx} (1+x^2y^2 - x)$$

$$\frac{dy}{dx} = \frac{y}{1+x^2y^2 - x}$$

Applications of Differentiation of Inverse Trigonometric Functions

68. Find the equations of the tangent and the normal to $y = [\sin^{-1} x]^2$ at the point where

$$x = \frac{-1}{\sqrt{2}}. \quad \text{68} \quad y = (\sin^{-1} x)^2 \text{ at the point } x = \frac{-1}{\sqrt{2}}$$

$$y' = 2(\sin^{-1} x) \cdot \frac{1}{\sqrt{1-x^2}}$$

$$x = \frac{-1}{\sqrt{2}}, m_T = 2 \left[\sin^{-1} \left(\frac{-1}{\sqrt{2}} \right) \right] \cdot \frac{1}{\sqrt{1 - \left(\frac{-1}{\sqrt{2}} \right)^2}}$$

$$= 2 \left(-\frac{\pi}{4} \right) \cdot \frac{1}{\left(\frac{1}{\sqrt{2}} \right)}$$

$$= -\frac{\pi}{2} \cdot \sqrt{2}$$

$$m_T = -\frac{\pi\sqrt{2}}{2}$$

$$m_N = -\frac{1}{m_T} = \frac{+1}{\frac{\pi\sqrt{2}}{2}} = \frac{2}{\pi\sqrt{2}} = \frac{\sqrt{2}}{\pi}$$

point: $x = \frac{-1}{\sqrt{2}}$
 $y = (\sin^{-1}(\frac{-1}{\sqrt{2}}))^2 = \left(-\frac{\pi}{4}\right)^2 = \frac{\pi^2}{16}$
 $\left(\frac{-1}{\sqrt{2}}, \frac{\pi^2}{16}\right)$

Equation of the tangent at $\left(\frac{-1}{\sqrt{2}}, \frac{\pi^2}{16}\right)$ is

$$y - \frac{\pi^2}{16} = -\frac{\pi\sqrt{2}}{2} \left(x + \frac{1}{\sqrt{2}}\right)$$

$$y = -\frac{\pi\sqrt{2}}{2}x - \frac{\pi}{2} + \frac{\pi^2}{16}$$

Equation of the Normal at $\left(\frac{-1}{\sqrt{2}}, \frac{\pi^2}{16}\right)$ is

$$y - \frac{\pi^2}{16} = \frac{\sqrt{2}}{\pi} \left(x + \frac{1}{\sqrt{2}}\right)$$

$$y = \frac{\sqrt{2}}{\pi}x + \frac{1}{\pi} + \frac{\pi^2}{16}$$

69. (a) Prove that $\frac{d}{dx} \left[\tan^{-1}(x) + \tan^{-1}\left(\frac{1}{x}\right) \right] = 0$ for $x > 0$, and hence show that

$$\tan^{-1}(x) + \tan^{-1}\left(\frac{1}{x}\right) = \frac{\pi}{2}, \text{ for } x > 0.$$

(b) Find the value of $\tan^{-1}(x) + \tan^{-1}\left(\frac{1}{x}\right)$ for $x < 0$.

69. (a) $\frac{d}{dx} \left[\tan^{-1}(x) + \tan^{-1}\left(\frac{1}{x}\right) \right] = 0$ for $x > 0$ (b) Find the Value of $\left[\tan^{-1}(x) + \tan^{-1}\left(\frac{1}{x}\right) \right]$ for $x < 0$

Let $f(x) = \tan^{-1}(x) + \tan^{-1}\left(\frac{1}{x}\right), x > 0$

$$f'(x) = \frac{1}{1+x^2} + \frac{\left(\frac{-1}{x^2}\right)}{1+\left(\frac{1}{x}\right)^2} \cdot \frac{x^2}{x^2}$$

$$= \frac{1}{1+x^2} - \frac{1}{x^2+1}$$

$$f'(x) = 0, x > 0 \Rightarrow \frac{d}{dx} \left[\tan^{-1}(x) + \tan^{-1}\left(\frac{1}{x}\right) \right] = 0$$

$\therefore f(x)$ is a constant when $x > 0$

$$x=1, f(1) = \tan^{-1}(1) + \tan^{-1}\left(\frac{1}{1}\right)$$

$$= \frac{\pi}{4} + \frac{\pi}{4}$$

$$= \frac{\pi}{2}$$

$$\therefore \tan^{-1}(x) + \tan^{-1}\left(\frac{1}{x}\right) = \frac{\pi}{2} \text{ for } x > 0$$

Let $f(x) = \tan^{-1}(x) + \tan^{-1}\left(\frac{1}{x}\right)$

Let $x = -y, y > 0$

$$f(-y) = \tan^{-1}(-y) + \tan^{-1}\left(\frac{1}{-y}\right)$$

$$= -\tan^{-1}(y) - \tan^{-1}\left(\frac{1}{y}\right)$$

$$= -\left[\tan^{-1}(y) + \tan^{-1}\left(\frac{1}{y}\right) \right], y > 0$$

$$f(-y) = -\frac{\pi}{2} \text{ (from (a))}$$

$$\therefore f(x) = -\frac{\pi}{2}$$

$$\therefore \tan^{-1}(x) + \tan^{-1}\left(\frac{1}{x}\right) = -\frac{\pi}{2}, x < 0.$$

Symmetric Properties

1. $\sin^{-1}(-x) = -\sin^{-1}x$
2. $\cos^{-1}(-x) = \pi - \cos^{-1}x$
3. $\tan^{-1}(-x) = -\tan^{-1}x$

Applications of Differentiation of Inverse Trigonometric Functions continued

70. (a) Show that the curve $y = x \sin^{-1} x$ has a stationary point at $(0,0)$ and determine its nature. You may assume that this is the only stationary point.
- (b) Show that the function $y = x \sin^{-1} x$ is even.
- (c) Find the domain and range of the function $y = x \sin^{-1} x$.
- (d) Hence sketch the graph of the function $y = x \sin^{-1} x$.

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(a)

$$y = \frac{x \sin^{-1} x}{1}$$

$$y'(x) = x \cdot \frac{1}{\sqrt{1-x^2}} + \sin^{-1} x \cdot 1$$

$$y'(x) = \frac{x}{\sqrt{1-x^2}} + \sin^{-1} x$$

Substitute $(0,0)$

$$y'(0) = \frac{0}{\sqrt{1-0^2}} + \sin^{-1} 0$$

$$y'(0) = 0 + 0$$

$$y'(0) = 0$$

$\therefore (0,0)$ is the stationary point.

Nature:

$$y'(x) = \frac{x}{\sqrt{1-x^2}} + \sin^{-1} x$$

x	$-\frac{1}{2}$	0	$\frac{1}{2}$
$y'(x)$	-	0	+

Minimum at $(0,0)$.

(b) $f(x) = x \sin^{-1} x$

$$\begin{aligned} f(-x) &= (-x) \sin^{-1}(-x) \\ &= -x (-\sin^{-1} x) \\ &= x \sin^{-1} x \end{aligned}$$

$$f(-x) = f(x)$$

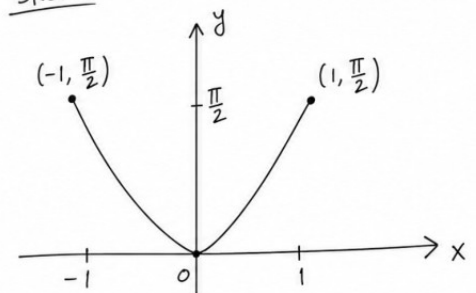
$\therefore f(x)$ is an even function.

(c) $y = x \sin^{-1} x$

Domain: $[-1, 1]$ ✓

Range: $x = -1, y = (-1) \sin^{-1}(-1) = \frac{\pi}{2}$
 $x = 1, y = 1 \cdot \sin^{-1}(1) = \frac{\pi}{2}$
 $x = 0, y = 0 \cdot \sin^{-1}(0) = 0$
 $[0, \frac{\pi}{2}]$ ✓

(d) sketch



Applications of Differentiation of Inverse Trigonometric Functions continued

71. A point P is observed on a vertical pole 10 metres above the level of an observer's eye. Q is a point 30 metres above P . If the observer is x metres from the pole, prove that the angle β subtended by PQ at the observer's eye is given by

$$\beta = \tan^{-1}\left(\frac{40}{x}\right) - \tan^{-1}\left(\frac{10}{x}\right).$$

Hence, or otherwise, find where the observer must stand in order for this angle β to be greatest.

$\beta = \alpha - \gamma$

$\tan \alpha = \frac{AQ}{OA} = \frac{40}{x}$
 $0 < \alpha < \frac{\pi}{2}$ } $\alpha = \tan^{-1}\left(\frac{40}{x}\right)$

$\tan \gamma = \frac{AP}{OA} = \frac{10}{x}$
 $0 < \gamma < \frac{\pi}{2}$ } $\gamma = \tan^{-1}\left(\frac{10}{x}\right)$

$\therefore \beta = \tan^{-1}\left(\frac{40}{x}\right) - \tan^{-1}\left(\frac{10}{x}\right) \checkmark$

$\beta(x) = \tan^{-1}\left(\frac{40}{x}\right) - \tan^{-1}\left(\frac{10}{x}\right)$

$\beta'(x) = \frac{\left(\frac{-40}{x^2}\right)}{1 + \left(\frac{40}{x}\right)^2} - \frac{\left(\frac{-10}{x^2}\right)}{1 + \left(\frac{10}{x}\right)^2}$

$\beta'(x) = \frac{-40}{1600 + x^2} + \frac{10}{100 + x^2}$

Stationary Points:

$\beta'(x) = 0$

$\frac{-40}{1600 + x^2} + \frac{10}{100 + x^2} = 0$

$\frac{40}{1600 + x^2} = \frac{10}{100 + x^2}$

$4(100 + x^2) = 1600 + x^2$

$400 + 4x^2 = 1600 + x^2$

$3x^2 = 1200$

$x^2 = 400$

$x = \pm 20$ [$\because x > 0$]

$x = 20\text{m} \checkmark$

Nature: $\beta'(x) = \frac{-40}{1600 + x^2} + \frac{10}{100 + x^2}$

x	10	20	30
$\beta'(x)$	+	0	-

$\beta(x)$ is maximum when $x = 20\text{m}$.

Maximum angle of β is

$x = 20$

$\beta(20) = \tan^{-1}\left(\frac{40}{20}\right) - \tan^{-1}\left(\frac{10}{20}\right)$

$\beta = \tan^{-1}(2) - \tan^{-1}\left(\frac{1}{2}\right)$

$\tan \beta = \tan\left[\tan^{-1}(2) - \tan^{-1}\left(\frac{1}{2}\right)\right]$

$= \frac{2 - \frac{1}{2}}{1 + 2 \cdot \frac{1}{2}}$

$= \frac{\frac{3}{2}}{2}$

$\tan \beta = \frac{3}{4}$ } $\beta = \tan^{-1}\left(\frac{3}{4}\right)$

$0 < \beta < \frac{\pi}{2}$

greatest angle of β

Applications of Differentiation of Inverse Trigonometric Functions continued

72. A plane maintains a constant altitude of 6000 metres whilst flying horizontally away from an airport at a constant speed of 600 km/hr. Let x km be the horizontal distance of the plane from the airport and let θ be the angle of elevation of the plane from the airport.

- (a) Show that $\theta = \tan^{-1} \frac{6}{x}$.
- (b) Hence find, in radians per second, the rate at which θ is decreasing when $x = 3$.

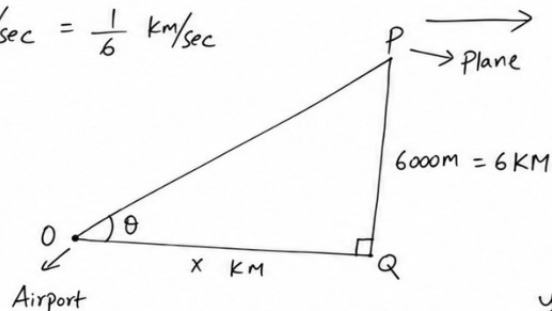
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Given $\frac{dx}{dt} = 600 \text{ km/hr} = \frac{600}{3600} \text{ km/sec} = \frac{1}{6} \text{ km/sec}$

(a)

$$\tan \theta = \frac{PQ}{OQ}$$

$$\left. \begin{array}{l} \tan \theta = \frac{6}{x} \\ 0 < \theta < \frac{\pi}{2} \end{array} \right\} \Rightarrow \theta = \tan^{-1} \frac{6}{x}$$



(b)

find $\frac{d\theta}{dt}$ when $x = 3 \text{ km}$

$$\begin{aligned} \theta &= \tan^{-1} \frac{6}{x} \\ \frac{d}{dt}(\theta) &= \frac{d}{dt} \left(\tan^{-1} \frac{6}{x} \right) \\ \frac{d\theta}{dt} &= \frac{-\frac{6}{x^2}}{1 + \left(\frac{6}{x}\right)^2} \cdot \frac{dx}{dt} \\ &= \frac{-6}{x^2 + 36} \cdot \frac{1}{6} \end{aligned}$$

$$\frac{d\theta}{dt} = \frac{-1}{x^2 + 36}$$

$$x = 3, \quad \frac{d\theta}{dt} = \frac{-1}{3^2 + 36} = \frac{-1}{45} \text{ rad/sec}$$

$\therefore \theta$ is decreasing at a rate of $\frac{1}{45} \text{ rad/sec}$

$$y = \tan^{-1} f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{1 + (f(x))^2}$$

$$y = \tan^{-1} x$$

$$-\frac{\pi}{2} < y < \frac{\pi}{2}$$